

Hybrid Hidden Markov Model for Modeling Equity Excess Growth Rate Dynamics: A Discrete-State Approach with Jump-Diffusion

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Generating synthetic financial time series that preserve the statistical properties of real market data is essential for stress testing, risk model validation, and scenario design. Existing approaches struggle to simultaneously reproduce heavy-tailed distributions, negligible linear autocorrelation, and persistent volatility clustering. We developed a hybrid hidden Markov framework that discretized excess growth rates into Laplace quantile-defined states and augmented regime switching with a Poisson jump-duration mechanism to enforce realistic tail-state dwell times. Parameters were estimated by direct transition counting, bypassing the Baum-Welch EM algorithm and scaling to a 424-asset pipeline. Applied to ten years of daily equity data, the framework achieved high distributional pass rates both in-sample and out-of-sample while partially reproducing the volatility clustering that standard regime-switching models miss. No single model was best at everything: GARCH(1,1) better reproduced volatility clustering but failed distributional tests, while the standard HMM without jumps passed more distributional tests but could not generate volatility clustering. The proposed framework delivered the most balanced performance overall. For multi-asset generation, copula-based dependence models that preserved each asset's marginal HMM distribution substantially outperformed a Single-Index Model factor baseline on both per-asset distributional accuracy and correlation reproduction.

Additional Key Words and Phrases: Synthetic Data Quality, Time Series Generation, Hidden Markov Model, Jump-Diffusion, Volatility Clustering, Regime-Switching, Stylized Facts, Distributional Fidelity

1 Introduction

Generating synthetic financial time series that faithfully preserve the statistical properties of real market data is a central challenge in quantitative finance and data quality research. High-fidelity synthetic data are essential for stress testing risk models against plausible but unobserved market scenarios, validating portfolio optimization algorithms, and augmenting limited training sets for machine learning pipelines [5, 35]. Yet the quality requirements for financial synthetic data are demanding: empirical equity excess growth rates exhibit three well-documented statistical regularities, namely heavy-tailed (leptokurtic) distributions, negligible linear autocorrelation in raw returns, and persistent volatility clustering, in which large changes tend to be followed by large changes of either sign and small changes by small changes [44], that any generative framework must simultaneously reproduce to be considered statistically faithful. These domain-specific quality criteria, often called *stylized facts*, serve as the benchmark against which the fidelity of any synthetic financial data must be evaluated (Figure 1) [17, 44, 55]. All three regularities are visible in SPY daily excess growth rates (Figure 1): the heavy-tailed distribution (1a), departures from normality in the tails (1b), near-zero autocorrelation of raw returns (1c), and the slow decay of absolute-return autocorrelation that characterizes volatility clustering (1d). The challenge of reproducing all three stylized facts simultaneously has been a long-standing problem in financial time series generation, and no single generative framework has yet succeeded in doing so at scale.

Existing generative approaches address these quality requirements from different angles but each has shortcomings. Generalized Autoregressive Conditional Heteroscedasticity (GARCH)-family models [11, 23] capture conditional variance dynamics but do not explicitly represent discrete market

regimes or sudden jumps in volatility. Stochastic-volatility and jump-diffusion models [33, 45] enrich tail behavior but lack a mechanism for volatility persistence after jumps. Deep generative models, including recurrent neural networks [28, 34] and generative adversarial networks [40, 56, 57], can learn complex distributions from data but struggle to reproduce temporal dependence, particularly volatility clustering, unless the network architecture is specifically designed to capture them [55]. Hidden Markov models (HMMs) and related regime-switching frameworks offer a probabilistic structure for non-stationary market behavior through latent state dynamics [47], yet standard HMMs fail to generate persistent high-volatility regimes, leading to reversion following extreme events and poor temporal quality in the synthetic output [12, 49]. Bridging this gap requires a generative framework that preserves both distributional fidelity and temporal structure while remaining computationally scalable.

In this study, we addressed this challenge by developing a hybrid HMM framework that discretized continuous excess growth rates into quantile-based regimes. The resulting Markov process was augmented with a two-parameter Poisson jump-duration mechanism that forced the model to dwell in high-volatility tail states for empirically realistic durations. By partitioning excess growth rates into quantile-defined states via a Laplace cumulative distribution function, we enabled direct frequentist counting of regime transitions, bypassing the iterative Baum-Welch algorithm (and its sensitivity to starting values) entirely, making the framework scalable to a 424-asset synthetic data pipeline. We evaluated synthetic data quality using five complementary metrics: Kolmogorov-Smirnov (KS) and Anderson-Darling (AD) pass rates for distributional fidelity, Wasserstein-1 and Hellinger distances to measure how far synthetic distributions deviated from the observed data, and autocorrelation function mean absolute error (ACF-MAE) for temporal structure preservation. Using a dataset of 424 United States-listed equities spanning ten years (2014–2024) for training and one year for out-of-sample testing, with the SPDR S&P 500 ETF Trust (SPY) as the primary validation asset, the framework achieved high distributional pass rates both in-sample and out-of-sample while partially reproducing the volatility clustering that standard regime-switching models miss. A Single-Index Model factor extension [51] propagated the univariate SPY engine to the full 424-asset universe, preserving cross-sectional correlation structure but producing poor distributional fits for assets that did not closely track the index. To address this limitation, we evaluated copula-based dependence models (Gaussian, Student-t, and C-vine) that preserved each asset’s fitted HMM marginal while injecting cross-asset correlation via rank reordering [1, 22]; the Student-t copula substantially outperformed the linear factor baseline on both per-asset distributional accuracy and correlation reproduction. Tunable hyperparameters controlled the jump frequency and duration, allowing users to trade off temporal and distributional quality for a given asset. No single model was best at everything: GARCH(1,1) better reproduced volatility clustering but failed distributional tests, and the standard HMM without jumps passed more distributional tests but could not generate volatility clustering. Thus, the proposed framework avoided the failures of each alternative and delivered the most balanced performance overall.

2 Related Work

Research on reproducing the canonical stylized facts of financial markets divides into bottom-up agent-based models, which derive return dynamics from simulated interactions of heterogeneous agents [4, 14, 18, 26, 27, 41, 43], and top-down data-driven models, which fit statistical structures directly to observed return series. While agent-based models produce stylized facts as emergent properties, they can be challenging to calibrate to specific assets and computationally demanding as practical generators. The data-driven tradition is the focus of the present review; the subsections below trace the evolution from parametric return models through latent-state and deep generative approaches to the multi-asset extensions that motivate this study.

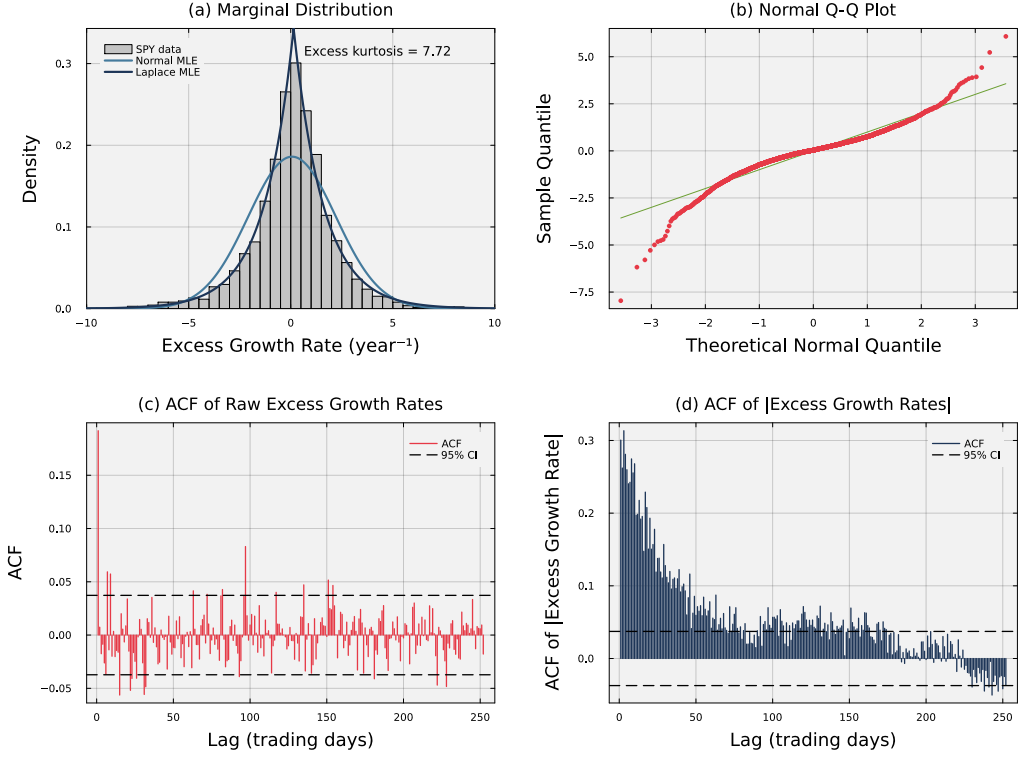


Fig. 1. Empirical stylized facts of SPY daily excess growth rates (2014–2024). Panel (a) shows the leptokurtic marginal distribution; the Laplace fit substantially outperforms the Gaussian. Panel (b) confirms heavy tails via a normal quantile-quantile (Q-Q) plot. Panels (c) and (d) contrast the near-zero autocorrelation of raw returns (consistent with the efficient markets hypothesis) against the persistent autocorrelation of absolute returns, motivating the jump-duration extension.

2.1 Stylized facts and the limitations of parametric models

The Black-Scholes-Merton option pricing model [10] and its underlying geometric Brownian motion (GBM) assume that log-returns are i.i.d. Gaussian: prices move smoothly, volatility is constant, and extreme moves are negligibly rare. These properties carry well-known practical advantages: closed-form option prices, simple portfolio variance calculations, and minimal data requirements. However, the Gaussian diffusion paradigm conflicts with a broad set of empirical regularities documented well before the model’s widespread adoption. Mandelbrot [44] demonstrated that speculative price changes exhibit heavy tails (*leptokurtosis*) beyond Gaussian predictions. Fama [24] reviewed evidence that raw return series display negligible linear autocorrelation, consistent with the efficient markets hypothesis. Yet Mandelbrot also observed that large price changes tend to cluster together regardless of sign, a pattern now called *volatility clustering*. Schwert [50] and others later confirmed this statistically, showing that the autocorrelation of absolute returns decays slowly over time. Cont [17] provides a systematic review confirming these regularities across asset classes, geographic markets, and sampling frequencies. Collectively, heavy tails (leptokurtosis), the absence of return predictability, and volatility clustering define the benchmark against which any model of market dynamics must be evaluated.

Engle [23] introduced autoregressive conditional heteroscedasticity (ARCH) models to address volatility clustering, and Bollerslev [11] generalized this to the GARCH(p,q) specification. GARCH models are widely used but have well-documented limitations. Heavy tails must be injected by choosing a fat-tailed noise distribution (e.g., Student-t); the model itself does not produce them. There is no representation of discrete market regimes or sudden jumps. Volatility persistence is controlled by a single decay parameter, so the model cannot distinguish between a calm market that slowly heats up and a sudden crash that lingers [20]. Stochastic volatility models [33, 54] treat volatility as a latent process, but require computationally intensive estimation. An alternative approach introduces Poisson-driven jumps directly into the price process. Merton [45] added a compound Poisson jump component to GBM, producing a Gaussian-mixture marginal distribution that directly addresses leptokurtosis, and Kou [39] extended this framework with asymmetric jump sizes to capture the empirical pattern that crashes tend to be sharper than rallies. Jump-diffusion models are particularly relevant in equity markets, where macroeconomic announcements, earnings surprises, and geopolitical events generate discrete large-magnitude price movements [6]. However, in their standard form, jump-diffusion models do not provide a mechanism for volatility persistence: the post-jump volatility immediately returns to its baseline level, absent a separate stochastic volatility component. Hidden Markov models offer a structurally different solution by embedding these dynamics in a latent discrete-state process.

2.2 Hidden Markov models and regime-switching

Hamilton [31] introduced the regime-switching HMM to economics, showing that U.S. GNP growth is well described by a two-state model with distinct expansion and recession regimes. Since then, HMMs have become a standard tool for latent-state modeling of financial time series, with applications ranging from volatility forecasting [48] to equity trading strategies [46]. Rydén, Teräsvirta, and Åsbrink [49] systematically evaluated HMMs against the stylized facts of daily returns. Gaussian-emission HMMs with few states reproduce heavy tails and the absence of return autocorrelation, but fail to capture volatility clustering: the ACF of absolute returns decays far too quickly relative to empirical data. Bulla and Bulla [12] showed that hidden semi-Markov models (HSMMs), which replace the geometric sojourn-time distribution with a flexible alternative, substantially improve volatility-clustering reproduction by allowing realistic persistence in high-volatility states. However, both standard and semi-Markov approaches rely on EM estimation (Baum-Welch [7]), which requires iterative forward-backward passes and is sensitive to initialization. The present study sidesteps this limitation by replacing EM with direct frequentist counting of transitions between quantile-defined states, an approach that is computationally trivial and initialization-free. Alongside HMM-based methods, a parallel line of work has explored whether deep generative models can learn stylized facts directly from data.

2.3 Synthetic data generation and deep generative models

Synthetic financial time series are increasingly used to augment limited data, stress test portfolios, and benchmark generative models. Assefa et al. [5] surveyed this landscape, noting that stylized-facts reproduction is the primary quality criterion and that naive generators tend to fail on at least one core stylized fact. Deep generative models, particularly generative adversarial networks (GANs), have attracted interest as a way to learn these properties directly from data, but results have been mixed. Takahashi, Chen, and Tanaka-Ishii [56] found that standard GAN architectures produced return sequences that visually resembled financial data yet failed to reproduce ACF structure, and Kwon and Lee [40] confirmed that volatility clustering remained elusive unless the architecture was specifically designed for it. Foundation models for Markov jump processes [8] offer a complementary pretrained-inference perspective, though their application to financial series

remains open. The hybrid HMM of the present study takes a different path: it replaces iterative EM with direct frequentist counting (eliminating training instability) and augments the Markov chain with a jump-duration mechanism that addresses the volatility-clustering gap without the architectural complexity of deep generative models. All of the models discussed so far are univariate; extending them to multi-asset portfolios introduces a separate set of challenges.

2.4 Factor models and multi-asset extensions

Existing HMM-based work in finance has been predominantly univariate [31, 36, 46, 48], leaving open the question of how HMM-derived generative models can scale to large cross-sections of assets. One strategy could be to use an HMM-based approach to capture the marginal distribution of a market factor (e.g., SPY) and then reconstruct individual asset paths via a linear factor model. The Single-Index Model (SIM) of Sharpe [51] decomposes each asset’s excess return into a common market factor and an idiosyncratic residual, reducing the parameter count relative to a full multivariate model. Mixture hidden Markov models [19] allow multiple assets to share a latent state structure, but joint estimation across hundreds of assets is computationally demanding. On the other hand, copulas offer an alternative by separating marginal distributions from the dependence structure [15, 22], though estimation grows expensive in high dimensions and the choice of copula family adds model selection complexity. The present study addresses the multi-asset challenge through both approaches: a SIM factor decomposition provides a scalable baseline for a 424-asset universe (the hybrid HMM generates market-factor paths from SPY, and asset-level paths are reconstructed via the linear projection), while copula-based dependence models (Gaussian, Student-t, and C-vine [1]) preserve each asset’s independently fitted HMM marginal and inject cross-asset dependence via rank reordering. The empirical comparison reveals that copulas substantially improve per-asset distributional accuracy relative to the SIM, and that a single Student-t copula is a parsimonious and sufficient specification for the equity pairs examined.

3 Methods

Data.

The empirical foundation of this study was a dataset composed of 424 United States-listed equities and exchange-traded funds spanning 10 years (2014-2024). A pipeline was developed to support the automated construction of excess growth rate models for any ticker within this dataset, allowing for batch simulation and stress testing across market sectors. To validate the performance of this pipeline, we focused our analysis on the single-asset SPY, which tracked the Standard & Poor's (S&P) 500 index. The data included daily open, high, low, and close prices along with volume-weighted average price metrics (VWAP). For training purposes, the first 2,766 observations constituted the in-sample dataset, spanning from January 3, 2014, to December 31, 2024. A subsequent 249 trading days, from January 2 to December 31, 2025, were reserved for out-of-sample testing to assess model generalizability.

Excess growth rate calculation.

We computed the excess growth rate $G_{i,j}$ for ticker i between time period $j - 1 \rightarrow j$ given the equity price series $P_{i,j}$:

$$G_{i,j} \equiv \left(\frac{1}{\Delta t} \right) \cdot \ln \left(\frac{P_{i,j}}{P_{i,j-1}} \right) - r_f \quad (1)$$

where $\Delta t = 1/252$ (one trading day in years) and r_f was a constant continuously compounded risk-free rate derived from STRIPS bond yields. The excess growth rate has units of year^{-1} and is time-additive under continuous compounding. A constant r_f was used to isolate volatility patterns without adding noise from interest rate movements.

Discrete hidden Markov model growth rate generator.

We defined the hidden Markov model as the tuple $\mathcal{M} = (\mathcal{S}, \mathcal{O}, \mathbf{T}, \mathbf{E}, \bar{\pi})$ [47], where the hidden state $S_t \in \mathcal{S} = \{1, \dots, N\}$ represented market regimes, the observation space $\mathcal{O} \subset \mathbb{R}$ consisted of continuous excess growth rate values, $\mathbf{T}$ governed regime-to-regime transitions, $\mathbf{E}$ specified the state-conditional emission model, and $\bar{\pi}$ denoted the stationary distribution. States were defined by quantile boundaries $Q_k = F_L^{-1}(k/N; \mu_L, b_L)$ of a fitted Laplace distribution, with $Q_0 = -\infty$ and $Q_N = +\infty$; the Laplace was chosen for its sharper peak (better matching the concentration of small price movements [17, 38, 44]) and its closed-form quantile function, which scaled to the 424-ticker pipeline without iterative optimization [9]. An observation was assigned to state k when $Q_{k-1} < G_t \leq Q_k$. The within-state emission followed a location-scale Student-t distribution:

$$G_t \mid S_t = k \sim \mu_k + \sigma_k \cdot t_\nu, \quad \nu = 5, \quad (2)$$

where $\nu = 5$ was selected from a sensitivity analysis over $\nu \in \{3, 4, 5, 6, 7, 8, 10, 15, 30, \infty\}$ as the value that minimized the kurtosis gap without degrading distributional fidelity ($\nu = \infty$ recovers the Normal baseline).

We estimated the transition matrix $\mathbf{T}$ through direct frequentist counting: the discrete state assignments allowed straightforward counting of observed transitions between regimes, making the approach conceptually simple and free of the initialization sensitivity inherent to EM [9]. A consequence of direct counting was that empirical transition matrices often had low probabilities for exiting central states into tail states, which caused models to lose the effect of volatility clustering too quickly [12]. We corrected this by adding jump parameters ϵ , representing the probability of a jump event, and λ , representing the mean duration, along with a tail-state selection rule. Tail-states were defined as $\mathcal{S}_{bottom} = \{1, \dots, N_{tail}\}$ and $\mathcal{S}_{top} = \{N - N_{tail} + 1, \dots, N\}$, where N_{tail} was user-configurable. If a jump was triggered, the jump-duration K was sampled from a Poisson

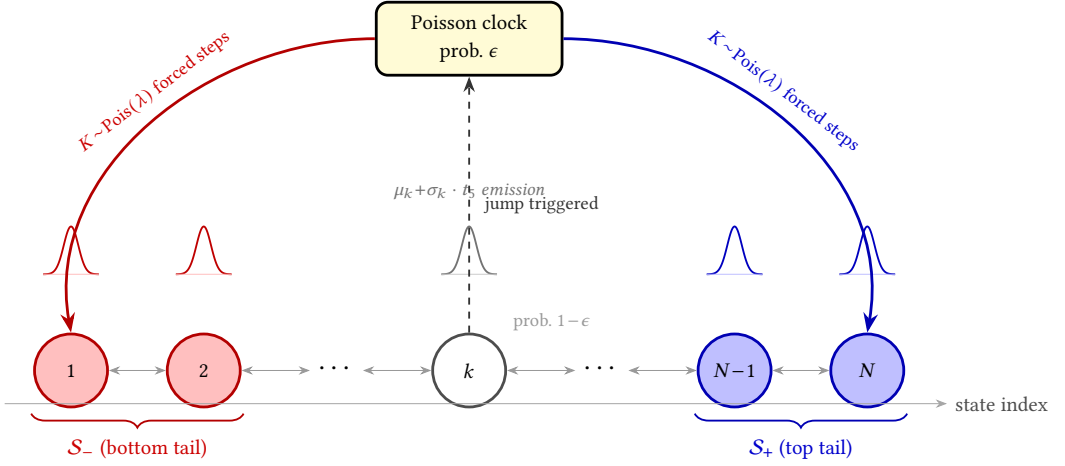


Fig. 2. Architecture of the Hybrid Hidden Markov Model with Jump-Diffusion (HMM-WJ). At each step the chain transitions according to the empirical transition matrix T (probability $1 - \epsilon$, thin bidirectional arrows) or enters a Poisson jump episode (probability ϵ , dashed upward arrow to the Poisson clock). During a jump episode the active state is forced to the bottom tail set S_- (red, lowest-valued states) or the top tail set S_+ (blue, highest-valued states) for $K \sim \text{Poisson}(\lambda)$ consecutive steps before normal Markovian evolution resumes. Small bell curves above each state depict the state-conditional location-scale Student-t ($\nu = 5$) emission distribution. Tail sets are defined as the N_{tail} lowest- and highest-quantile states under the Laplace quantile partition.

distribution such that $K \sim \text{Poisson}(\lambda)$, which is standard for modeling independent events in fixed intervals [29]. During these K steps, the standard Markovian transitions were overridden to target tail-states, with a user-configurable bias p_{neg} (default 0.52) that favored negative-tail states to reproduce gain/loss asymmetry (Figure 2).

Given the state assignments and transition matrix, the remaining model components were estimated directly from the data. The emission parameters μ_k and σ_k were the sample mean and standard deviation of observations assigned to state k . The empirical transition matrix exhibited a dominant near-diagonal band reflecting short-range regime persistence (Figure S1, Online Appendix S1). Critically, under pure Markovian dynamics the model exited extreme (tail) states after only 1–2 steps on average, far shorter than the multi-week volatile episodes observed in real markets. This rapid reversion motivated the jump-duration extension. The stationary distribution $\bar{\pi}$ satisfied $\bar{\pi} = \bar{\pi}T$ and was estimated by matrix powering (T^{50}) [30, 32]. An initial state was drawn from $X_0 \sim \bar{\pi}$; at each subsequent step the chain either transitioned according to the empirical row $T_{X_{n-1}, \cdot}$ (probability $1 - \epsilon$) or entered a Poisson jump episode (probability ϵ) that forced the state into the tail set S_- or S_+ for $K \sim \text{Poisson}(\lambda)$ consecutive steps before normal transitions resumed. In both cases, an observation was sampled from the emission distribution $O_n \sim E_{X_n}$. The full simulation algorithm and remaining pipeline stages (model construction, state decoding) are given in Online Appendix S4 (Algorithm 1).

Hyperparameter optimization via grid search.

To identify the optimal jump probability (ϵ) and mean jump duration (λ) [39, 45], we implemented a multi-objective grid search that minimized the discrepancy between historical and simulated market signatures. The error function targeted two stylized facts [17, 49]: volatility clustering, measured

as the sum of squared errors between the observed and simulated autocorrelation function (ACF) of absolute excess growth rates [50] up to a lag of $L = 252$ trading days (approximately one trading year); and leptokurtosis, captured by a penalty term on the global kurtosis [44].

We defined the objective function $J(\epsilon, \lambda)$ as:

$$J(\epsilon, \lambda) = \sum_{\tau=1}^L \left(\text{ACF}_{obs}(\tau) - \overline{\text{ACF}}_{sim}(\tau) \right)^2 + w_K \left(K_{obs} - \bar{K}_{sim} \right)^2 \quad (3)$$

where $\text{ACF}_{obs}(\tau)$ and K_{obs} represented the empirical absolute growth autocorrelation at lag τ and the global kurtosis, respectively. The simulated equivalents, $\overline{\text{ACF}}_{sim}(\tau)$ and $\bar{K}_{sim}$, were averaged across 200 independent synthetic paths of 2,766 trading days per grid point [29]. We set the kurtosis penalty weight to $w_K = 0.20$ to balance tail behavior against the temporal ACF term. The grid search explored ϵ values from 10^{-4} to 2.5×10^{-2} and λ values from 10 to 160. The full computational procedure is detailed in Algorithm 4.

Validation metrics

Each of the 1,000 simulated paths was compared against the historical excess growth rate series using four complementary metrics. The two-sample Kolmogorov-Smirnov (KS) test [37, 52] assessed overall distributional agreement; the two-sample Anderson-Darling (AD) test [2] placed greater weight on the tails. For each test, we reported the *pass rate*: the proportion of paths for which the null hypothesis of distributional equivalence was not rejected at $\alpha = 0.05$. Both tests assume i.i.d. observations; because volatility clustering reduces the effective sample size, reported p-values may be slightly miscalibrated. To complement these binary verdicts with continuous effect-size measures, we also computed the Wasserstein-1 distance (W_1), which quantifies the minimum transport cost between two distributions in the same units as the data, and the Hellinger distance ($H \in [0, 1]$), which measures distributional overlap on a bounded scale. Both W_1 and H were averaged path-by-path across the 1,000 simulations; standard errors were estimated as $\text{std}/\sqrt{n_{\text{paths}}}$. Formal definitions of all four metrics are given in Supplemental Section S11.

Temporal fidelity metric

While KS and AD tests assess marginal distributional quality, they are insensitive to temporal dependence. To measure how well a generator reproduced volatility clustering, we defined the autocorrelation mean absolute error (ACF-MAE) as:

$$\text{ACF-MAE} = \frac{1}{L} \sum_{\tau=1}^L \left| \hat{\rho}_{|G|}^{\text{obs}}(\tau) - \hat{\rho}_{|G|}^{\text{sim}}(\tau) \right| \quad (4)$$

where $\hat{\rho}_{|G|}^{\text{obs}}(\tau)$ and $\hat{\rho}_{|G|}^{\text{sim}}(\tau)$ are the sample autocorrelations of the absolute excess growth rate series $|G_t|$ at lag τ for the observed and simulated paths, respectively, and $L = 252$ (one trading year). An i.i.d. generator produces $\hat{\rho}_{|G|}^{\text{sim}}(\tau) \approx 0$ for all $\tau > 0$, so its ACF-MAE equals the mean level of the observed autocorrelation function; lower values indicate better reproduction of empirical volatility persistence.

Multi-asset extensions

To generalize synthetic path generation beyond a single asset, we considered two dependence specifications. The Single-Index Model (SIM) [51] decomposes each asset's excess growth rate as $G_{i,t} = \alpha_i + \beta_i \cdot G_{\text{SPY},t} + \eta_{i,t}$, where β_i measures systematic sensitivity to the index and $\eta_{i,t}$ is an idiosyncratic shock. The parameters (α_i, β_i) were estimated via OLS, and synthetic multi-asset paths were generated by (i) producing a single HMM-generated SPY path, (ii) resampling the

empirical residuals $\hat{\eta}_{i,t}$ with replacement, and (iii) recovering $\hat{G}_{i,t} = \hat{\alpha}_i + \hat{\beta}_i \cdot \hat{G}_{SPY,t} + \hat{\eta}_{i,t}$. This construction preserved cross-sectional correlation via the common factor while maintaining the regime-switching dynamics of the index across all 424 assets.

The SIM imposes a linear, single-factor structure that cannot capture nonlinear or tail dependence. As an alternative, we considered copula-based dependence specifications [15, 22], which fit an independent HMM to each asset, simulate each path independently (preserving marginals exactly), and then *reorder* simulated values so that cross-asset dependence matches the historical structure. We evaluated three families: a *Gaussian copula* (zero tail dependence), a *Student-t copula* (symmetric tail dependence controlled by ν), and a *C-vine copula* [1] that decomposed the d -dimensional structure into a hierarchy of bivariate copulas selected per-edge by AIC from five candidate families. Copula parameters were estimated from Kendall’s rank correlation after converting growth rates to uniform marginals via the probability integral transform. Full mathematical details are given in Supplemental Section S6.

Table 1. Descriptive statistics and stylized-facts tests for SPY daily excess growth rates. In-sample: 2014–2024 ($T = 2,766$); out-of-sample: 2025 ($T = 249$). Gaussian and Laplace columns show maximum likelihood estimation (MLE) fits to the in-sample data. JB = Jarque-Bera normality test. LB = Ljung-Box autocorrelation test at lag 20. * denotes rejection at $\alpha = 0.05$.

Statistic	IS Observed	OoS Observed	Gaussian	Laplace
Mean (annualized, %)	6.31	11.60	6.31	14.19
Std Dev (annualized, %)	214.50	220.62	214.46	205.93
Skewness	-0.753	-1.093	0	0
Excess Kurtosis	7.715	6.867	0	3
JB normality test	reject* ($p < 0.001$)	reject* ($p < 0.001$)	n/a	n/a
LB test on G_t (lag 20)	reject* ($p < 0.001$)	reject* ($p = 0.019$)	n/a	n/a
LB test on $ G_t $ (lag 20)	reject* ($p < 0.001$)	reject* ($p < 0.001$)	n/a	n/a

4 Empirical Study

Descriptive statistics confirmed that all three canonical stylized facts were present in both the in-sample (2014–2024, $T = 2,766$) and out-of-sample (2025, $T = 249$) windows (Table 1): excess kurtosis of 7.7 (IS) and 6.9 (OoS) with Jarque-Bera rejection ($p < 0.001$), persistent volatility clustering (Ljung-Box on $|G_t|$ rejected, $p < 0.001$), and near-zero linear autocorrelation in raw returns [24].

4.1 Jump hyperparameter optimization and state resolution

With the stylized facts confirmed in both evaluation windows, we turned to calibrating the jump mechanism. A multi-objective grid search over $\epsilon \in [10^{-4}, 2.5 \times 10^{-2}]$ and $\lambda \in [10, 160]$, with 200 simulated paths per grid point, identified the optimal values $\epsilon^* = 10^{-4}$ and $\lambda^* = 100$ (Figure S2, Online Appendix S2). The optimal $\lambda^* = 100$ was an interior point of the λ grid, confirming that the mean jump duration was identifiable from data; $\epsilon^* = 10^{-4}$ fell at the lower boundary of the ϵ grid, indicating that the data consistently preferred rarer tail-entry events over the range explored. At these optimal values, the ACF of $|G_t|$ closely tracked the observed SPY ACF across lags 1–252, confirming that the jump mechanism partially reproduced the empirical ARCH effect.

The primary analysis used $N = 100$ states, but a sensitivity study over $N \in \{30, 60, 90, 100, 150, 200\}$ showed that the framework was robust to the choice of state resolution across a wide range (Table T1, Online Appendix S3). KS and AD pass rates remained stable for both the HMM without jumps (HMM-NJ; $\geq 98.9\%$ KS, $\geq 97.9\%$ AD) and HMM-WJ ($\geq 96.0\%$ KS, $\geq 88.6\%$ AD), while ACF-MAE for HMM-WJ improved slightly from 0.057 at $N = 30$ to 0.049 at $N = 200$. State-occupancy diagnostics confirmed that every state at $N \leq 200$ contained at least 7 observations, ensuring adequate statistical support per state under the Laplace equal-probability partition. At $N = 350$, however, certain states were never visited in the historical record, leaving zero transition probabilities that prevented the model from reaching all states, which established a practical upper bound on state resolution.

4.2 In-sample distributional and temporal quality

Having established that the model’s hyperparameters were identifiable and its state resolution robust, we benchmarked it against seven alternative generators. We generated 1,000 synthetic paths of 2,766 trading days from each of eight generators and evaluated them against the observed SPY series using two-sample KS and AD tests at $\alpha = 0.05$, along with the Wasserstein-1 distance, Hellinger distance, kurtosis matching, ACF fidelity, and quantile coverage (Table 2). We included six baselines to establish upper and lower bounds on expected quality (Table 2). Bootstrap resampling

set the non-parametric ceiling (100% KS, 100% coverage). The Gaussian i.i.d. baseline (0% KS) confirmed that thin-tailed models were categorically inadequate; Laplace recovered 44% KS by matching tail decay but missed regime-dependent variation. The HSMM baseline [12] achieved 82% KS with $K = 8$ states and negative-binomial dwell times, but empirical dwell times averaged only 1.1–1.8 steps, causing the semi-Markov structure to collapse to memoryless behavior. The gated recurrent unit (GRU) neural baseline (37,954 parameters) closed 83% of the ACF-MAE gap but catastrophically failed distributional tests (0.6% KS) due to variance collapse that reduced simulated standard deviation by 30%. This pattern illustrates the difficulty of simultaneously capturing both distributional and temporal quality dimensions with a single paradigm.

Turning from the baselines to the structured models, we evaluated KS/AD pass rates and kurtosis matching (Table 2, Figure 3, panels a and c). GARCH(1,1) achieved IS KS and AD pass rates of only 5.5% (SE 0.7) and 1.9% (SE 0.4), less than 6% of the Bootstrap distributional ceiling, which indicated that the GARCH marginal distribution was systematically inconsistent with the observed data despite its well-known volatility clustering properties. The HSMM occupied a middle ground (82.0% KS, 42.5% AD), outperforming GARCH and Laplace but falling well short of the HMM variants; its coarse 8-state partition could not capture the full shape of the growth rate distribution, as reflected by a simulated kurtosis of 4.8 against an observed 7.7 (38% gap) and only 68.7% quantile coverage. HMM-NJ achieved 99.7% (SE 0.2) and 99.1% (SE 0.3), and HMM-WJ achieved 97.6% (SE 0.5) and 91.3% (SE 0.9). Both HMM variants adequately captured the observed heavy-tailed distribution, though HMM-WJ's pass rates were 2–8 percentage points lower than HMM-NJ's, a cost of the jump mechanism which occasionally distorted the marginal distribution by overweighting tail states during jump episodes. Simulated excess kurtosis told a complementary story: the observed value was 7.7, GARCH overshot at 8.2 (SE 0.37) with high variance across paths, while HMM-NJ and HMM-WJ produced 8.1 (SE 0.14) and 7.6 (SE 0.14) respectively. The Student- $t(\nu = 5)$ emissions closely matched the observed kurtosis (within 1.3% for HMM-WJ), a substantial improvement over Normal emissions which underestimated kurtosis by approximately 29%. Skewness, which is often neglected in generative model evaluation, was also well reproduced: the observed skewness of -0.75 was matched by HMM-NJ (-0.76 , SE 0.01) and HMM-WJ (-0.73 , SE 0.01), while GARCH, Gaussian, and Laplace all produced near-zero skewness because their innovations are symmetric by construction. The HMM's ability to reproduce negative skewness arose from the asymmetric state-occupancy distribution inherited from the Laplace partition, not from any explicit skewness parameter; a sensitivity analysis over $\nu \in \{3, \dots, 30, \infty\}$ confirmed that $\nu = 5$ minimized the kurtosis gap without degrading other metrics. Quantile coverage reinforced these findings: Bootstrap, HMM-NJ, and HMM-WJ all achieved 100% coverage of the 1st–99th empirical percentiles, while Gaussian covered only 13.1%, Laplace 37.4%, GARCH 29.3%, and HSMM 68.7%, which indicated systematic tail misspecification in all non-HMM parametric generators.

Pass rates are binary verdicts; we complemented them with the Wasserstein-1 distance W_1 and Hellinger distance H ($H \in [0, 1]$, where 0 is identical and 1 is completely disjoint), which measure how far synthetic distributions differ from the observed data, independent of sample size (Table 2). In-sample, HMM-NJ and HMM-WJ stayed within 6% and 12% of the Bootstrap ceiling ($W_1 = 0.081$ and 0.101 vs. 0.062), while GARCH fell 72% of the way toward the Gaussian worst case ($W_1 = 0.304$). Out-of-sample, GARCH's W_1 deteriorated by 67% to 0.507 , worse than every model including Gaussian ($W_1 = 0.452$), despite recovering to 80.3% KS; this recovery was driven by reduced test power at $T = 249$ rather than genuine fidelity. By contrast, HMM-NJ and HMM-WJ maintained the lowest W_1 among parametric models in both windows (OoS $W_1 = 0.258$ and 0.282), confirming robust generalization.

The metrics above assess marginal distributions but are blind to temporal structure. We measured temporal quality using the mean absolute error of ACF($|G_t|$) over lags 1–252 (Figure 3, panel b).

Table 2. In-sample and out-of-sample model comparison for SPY (1,000 simulated paths, significance level $\alpha = 0.05$). Bootstrap: i.i.d. resample of training data. Gaussian/Laplace: i.i.d. draws from MLE-fitted distributions. GARCH: GARCH(1,1) estimated by quasi-maximum likelihood. GRU: 2-layer gated recurrent unit [16] with Gaussian output head (autoregressive generation). HSMM: hidden semi-Markov model with $K = 8$ Laplace quantile states, negative-binomial dwell times, and Student- $t(\nu = 5)$ emissions [12]. HMM-NJ: hidden Markov model ($N = 100$ states) without jumps. HMM-WJ: hybrid model with Poisson jump-duration extension. ACF-MAE: mean absolute error of the ACF of $|G_t|$ over lags 1–252. Wasserstein-1: mean absolute difference of sorted empirical quantiles (Eq. S14); units are daily excess growth rates. Hellinger: histogram-based distributional overlap distance in $[0, 1]$ (Eq. S15); $H = 0$ is identical, $H = 1$ is disjoint. Coverage: fraction of 99 empirical quantiles (1st–99th) falling within the [5th, 95th] percentile envelope of the corresponding synthetic quantiles. Standard errors in parentheses: binomial SE for pass rates, $\text{std}/\sqrt{n}$ for kurtosis, Wasserstein-1, and Hellinger, bootstrap SE ($B = 500$) for ACF-MAE and coverage. Arrows indicate preferred direction: $\uparrow$ higher is better, $\downarrow$ lower is better, $\approx$ closer to observed is better.

Metric	Bootstrap	Gaussian	Laplace	GARCH(1,1)	GRU	HSMM	HMM-NJ	HMM-WJ
<i>In-sample: 2,766 trading days (2014–2024)</i>								
KS pass rate (%) $\uparrow$	100.0 (<0.1)	0.0 (<0.1)	44.0 (1.6)	5.5 (0.7)	0.6 (0.2)	82.0 (1.2)	<u>99.7</u> (0.2)	97.6 (0.5)
AD pass rate (%) $\uparrow$	99.7 (0.2)	0.0 (<0.1)	43.3 (1.6)	1.9 (0.4)	0.2 (0.1)	42.5 (1.6)	<u>99.1</u> (0.3)	91.3 (0.9)
Excess kurtosis (observed)				7.715				
Excess kurtosis (simulated) $\approx$	7.6 (0.08)	−0.0 (<0.01)	3.0 (0.02)	<u>8.2</u> (0.37)	5.7 (0.16)	4.8 (0.08)	8.1 (0.14)	7.6 (0.14)
ACF-MAE $\downarrow$	0.060 (<0.001)	0.060 (<0.001)	0.060 (<0.001)	<u>0.031</u> (0.001)	0.036 (<0.001)	0.059 (<0.001)	0.059 (<0.001)	0.052 (<0.001)
Coverage (%) $\uparrow$	100.0 (<0.1)	13.1 (0.1)	37.4 (1.5)	29.3 (1.4)	17.2 (0.5)	68.7 (1.1)	<u>100.0</u> (<0.1)	<u>100.0</u> (<0.1)
Wasserstein-1 $\downarrow$	0.062 (0.001)	0.399 (0.001)	<u>0.138</u> (0.001)	0.304 (0.006)	0.421 (0.003)	0.176 (0.001)	<u>0.081</u> (0.001)	0.101 (0.001)
Hellinger dist $\downarrow$	0.042 (<0.001)	0.148 (<0.001)	<u>0.072</u> (<0.001)	0.100 (0.001)	0.134 (0.001)	0.113 (<0.001)	0.073 (<0.001)	0.075 (<0.001)
<i>Out-of-sample: 249 trading days (2025)</i>								
KS pass rate (%) $\uparrow$	97.4 (0.5)	62.2 (1.5)	88.0 (1.0)	80.3 (1.3)	71.8 (1.4)	96.2 (0.6)	<u>96.7</u> (0.6)	94.4 (0.7)
AD pass rate (%) $\uparrow$	98.5 (0.4)	46.3 (1.6)	92.9 (0.8)	72.0 (1.4)	44.8 (1.6)	<u>96.7</u> (0.6)	<u>96.7</u> (0.6)	95.1 (0.7)
Excess kurtosis (observed)				6.867				
Excess kurtosis (simulated) $\approx$	6.0 (0.18)	−0.0 (<0.01)	2.7 (0.05)	1.6 (0.07)	2.9 (0.10)	4.1 (0.13)	<u>6.4</u> (0.22)	6.1 (0.13)
ACF-MAE $\downarrow$	0.043 (<0.001)	0.043 (<0.001)	0.043 (<0.001)	<u>0.026</u> (<0.001)	0.033 (<0.001)	0.042 (<0.001)	0.041 (<0.001)	0.039 (<0.001)
Coverage (%) $\uparrow$	100.0 (<0.1)	36.4 (1.9)	74.7 (1.0)	96.0 (1.3)	77.8 (2.9)	<u>100.0</u> (0.2)	<u>100.0</u> (<0.1)	<u>100.0</u> (<0.1)
Wasserstein-1 $\downarrow$	0.232 (0.002)	0.452 (0.002)	0.263 (0.002)	0.507 (0.018)	0.488 (0.005)	0.287 (0.002)	<u>0.258</u> (0.002)	0.282 (0.006)
Hellinger dist $\downarrow$	0.205 (0.001)	0.235 (0.001)	0.211 (0.001)	0.232 (0.001)	0.240 (0.001)	0.239 (0.001)	<u>0.207</u> (0.001)	0.210 (0.001)
Parameters estimated [†]	0	2	2	3	37,954	18	2	4

The i.i.d. generators (Bootstrap, Gaussian, Laplace) all produced an ACF-MAE of 0.060, which defined the floor of no temporal structure; GARCH achieved the best-in-class ACF-MAE of 0.031, defining the ceiling. All other models fell between these anchors. GARCH’s advantage came at the cost of distributional fidelity (5.5% KS), a tradeoff the quality metrics exposed directly. HMM-NJ scored 0.059, closing only 3% of the gap between the i.i.d. floor and GARCH, which confirmed that the standard transition matrix alone was structurally incapable of generating persistent volatility clustering. The GRU achieved 0.036, closing 83% of the gap, demonstrating that the recurrent architecture effectively captured short-range temporal dependencies; however, this came at the cost of catastrophic distributional failure (0.6% KS, less than 1% of the Bootstrap ceiling). HMM-WJ scored 0.052, closing 28% of the gap, the only non-GARCH, non-neural model to reduce ACF-MAE meaningfully below the i.i.d. floor. The improvement arose because approximately 24% of HMM-WJ paths contained at least one Poisson jump event that sustained $\text{ACF}(|G_t|)$ decay across all lags, while the remaining 76% behaved identically to HMM-NJ. Importantly, the jump frequency is continuously tunable. Increasing ϵ produces more jump-containing paths and further reduces ACF-MAE, at the cost of degrading distributional fidelity. The grid search selected $\epsilon^* = 10^{-4}$ because this value minimized the joint objective over both temporal and distributional quality (Equation 3), meaning the 24% jump rate was the empirically optimal operating point for SPY, not a structural ceiling. The jump-duration mechanism was therefore the structural feature responsible for partially reproducing the ARCH effect, with its strength directly controlled by ϵ .

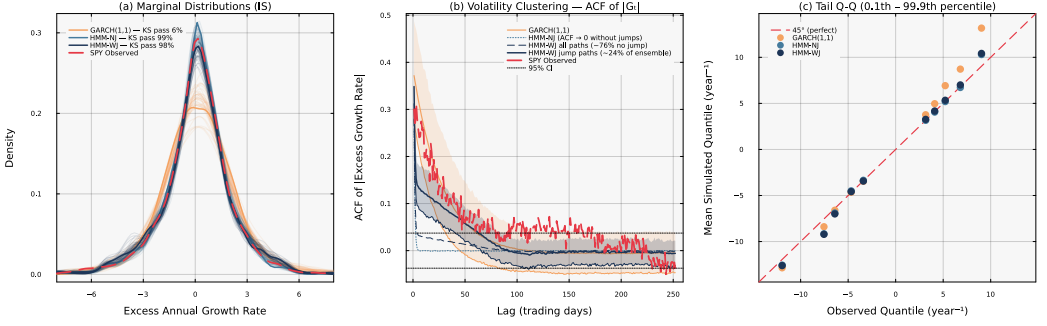


Fig. 3. Head-to-head in-sample model comparison for SPY ($N = 100, 1,000$ simulated paths). Panel (a): marginal density of excess growth rates with IS KS pass rates annotated. GARCH(1,1) fails the two-sample KS test on 95% of paths (pass rate 5.5%), indicating a systematic shape mismatch with the empirical distribution. Both HMM variants pass at rates $\geq 97\%$, confirming that the Student-t emission structure adequately captures the observed heavy tails. Out-of-sample, HMM-WJ maintains 94% KS pass rate, though HMM-NJ achieves a higher 97%; GARCH recovers partially (OoS 80%). Panel (b): autocorrelation function of absolute excess growth rates, $\text{ACF}(|G_t|)$, at lags 1–252 (shaded bands: 10th–90th percentile across paths). HMM-NJ is structurally incapable of producing persistent volatility clustering: without a jump mechanism the latent states are i.i.d. conditional on the Markov chain, so $\text{ACF}(|G_t|) \approx 0$ for all lags beyond one (dotted curve). HMM-WJ generates a *mixture* of two path families under the same model: approximately 76% of IS paths contain no Poisson jump and therefore behave identically to HMM-NJ (dashed curve, near zero); the remaining $\sim 24\%$ of paths contain at least one jump event and sustain substantial $\text{ACF}(|G_t|)$ decay across all lags (solid navy curve $\pm$ band). The jump frequency, and hence the fraction of the ensemble exhibiting slow ACF decay, is controlled by the tail-entry probability ϵ , making the volatility-clustering strength directly tunable. Panel (c): tail Q-Q plot at the 0.1st–99.9th percentile region; HMM-WJ provides the closest mean quantile match in the extreme tails.

4.3 Out-of-sample evaluation and model tradeoffs

The in-sample results demonstrated that HMM-WJ was the only model to improve on both distributional and temporal dimensions; the out-of-sample window tested whether this advantage survived on unseen data. We evaluated all eight generators on a held-out window of 249 trading days (full calendar year 2025) using the same metrics (Table 2, Figure 4). No single model dominated all quality dimensions simultaneously. GARCH(1,1) achieved the best temporal fidelity (closing 100% of the ACF-MAE gap) but the worst distributional fit among parametric models (5.5% of the Bootstrap KS ceiling). The GRU closed 83% of the ACF gap but achieved the worst distributional fit overall (0.6% of the KS ceiling), demonstrating that even 38,000 trainable parameters could not jointly capture both quality dimensions. HMM-NJ achieved the highest distributional pass rates (99.7% of the KS ceiling) but could not reproduce volatility clustering (closing only 3% of the ACF gap). HMM-WJ occupied the Pareto frontier: it closed 28% of the ACF gap while maintaining distributional pass rates above 91% of the ceiling, the only model to improve meaningfully on both dimensions simultaneously.

Out-of-sample, both HMM variants generalized robustly: HMM-NJ achieved 96.7% KS and HMM-WJ achieved 94.4% KS, with simulated kurtosis of 6.4 and 6.1 against an observed 6.9 (Table 2, OoS panel). GARCH’s W_1 deteriorated by 67% from IS to OoS, exposing a tail misspecification that binary

¹The $N \times N$ transition matrix in HMM-NJ and HMM-WJ is computed by direct counting of observed state transitions; it is a sufficient statistic of the data, not a fitted parameter. The counts listed here reflect only the scalar parameters requiring estimation (MLE or grid search).

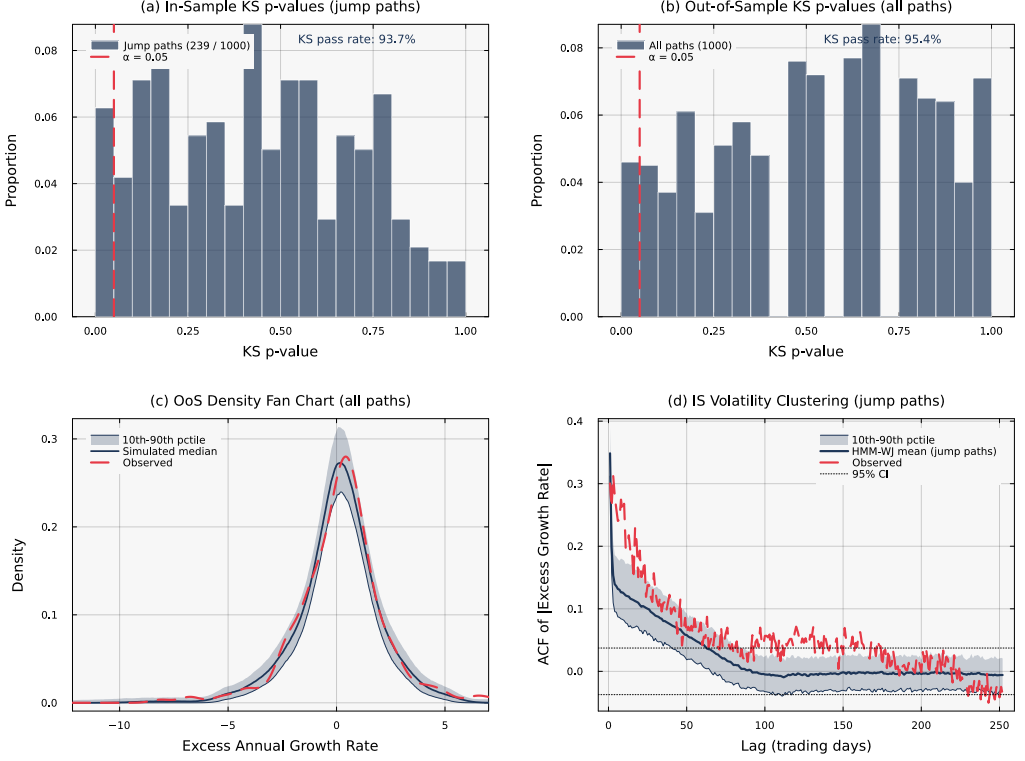


Fig. 4. Statistical validation of HMM-WJ ($N = 100$, $\alpha = 0.05$). Panels (a) and (d) condition on the $\sim 24\%$ of IS paths containing jump events. (a) IS KS p -values for 239 jump paths (pass rate 93.7%). (b) OoS KS p -values for all 1,000 paths (pass rate 95.4%). (c) OoS density fan chart; observed density falls within the simulation envelope. (d) IS ACF of $|G_t|$ for jump paths; the mean closely tracks the observed ACF. The all-paths vs. jump-paths ACF comparison is in Online Appendix S7.

pass rates obscured. The OoS ordering reversed the IS pattern (HMM-NJ now outperformed HMM-WJ), suggesting mild overfitting of the IS-calibrated jump parameters. Cross-asset generalization to NVDA, JNJ, and JPM confirmed that the jump mechanism automatically adapted its duration λ^* to each asset's clustering intensity (Online Appendix S5, Table T2).

4.4 Multi-asset extension

The single-asset results confirmed distributional and temporal fidelity for SPY; the remaining question was whether the framework could scale to a full asset universe. We propagated HMM-WJ SPY paths through a Single-Index Model (SIM) for 424 S&P 500 constituents, with idiosyncratic shocks drawn by resampling empirical residuals. The SIM yielded a median IS KS pass rate of 66.7% and a mean of 58.4% across the asset universe (well below the 95% expected under a correct model); high- β assets closely tracking SPY retained pass rates above 80%, while degradation concentrated among low- R^2 assets, confirming that the HMM generative engine was not the bottleneck. Out-of-sample, the mean KS pass rate reached 82.1% across 417 surviving assets. Full cross-sectional results, sector-level analysis, and the SIM R^2 distribution are reported in Online Appendix S8 (Figure S4, Table T3). To determine whether richer dependence specifications could recover the fidelity lost

Table 3. Dependence model comparison for a four-asset portfolio (SPY, NVDA, JNJ, JPM). Each model uses per-asset HMM marginals ($N = 100$, Student-t emissions with $\nu = 5$); only the dependence structure differs. 1,000 simulated paths, $\alpha = 0.05$. Corr Frobenius: root-sum-squared error of upper-triangle correlation entries (observed vs. mean simulated). Pairwise corr MAE: mean absolute error across all six asset pairs. SIM results exclude SPY (market factor).

	SIM	Gaussian	Student-t	Vine
<i>Marginal KS pass rate (%)</i>				
SPY	n/a	98.2 (0.4)	98.9 (0.3)	96.3 (0.6)
NVDA	79.5 (1.3)	96.6 (0.6)	99.1 (0.3)	98.6 (0.4)
JNJ	92.1 (0.9)	98.3 (0.4)	97.3 (0.5)	99.2 (0.3)
JPM	27.5 (1.4)	97.8 (0.5)	95.1 (0.7)	97.4 (0.5)
<i>Cross-asset dependence quality</i>				
Corr Frobenius error ↓	n/a	0.168 (0.001)	0.134 (0.001)	0.287 (0.002)
Pairwise corr MAE ↓	n/a	0.055	0.039	0.099

by the SIM’s linear assumption, we compared four dependence models on a four-asset portfolio (SPY, NVDA, JNJ, JPM): (i) a Gaussian copula, (ii) a Student-t copula, (iii) a C-vine copula with AIC-selected bivariate families, and (iv) the Single-Index Model with bootstrap residuals (Figure S5). We fitted an independent HMM to each asset, fitted each dependence model to the in-sample growth rate matrix, generated 1,000 synthetic paths of 2,766 steps, and evaluated both marginal fidelity (per-asset KS pass rates) and cross-asset dependence quality (correlation reproduction). For the copula models, simulation proceeded by rank reordering: each asset’s independently simulated HMM values were rearranged in time so that the cross-asset rank structure matched the fitted copula, preserving each marginal distribution exactly while injecting the desired correlation and tail dependence (Online Appendix S10).

We observed a clear separation between copula-based and factor-based approaches on marginal fidelity (Table 3, Figure 5, panel a). All three copula specifications achieved KS pass rates above 95% for every ticker, which confirmed that the rank-reordering mechanism preserved each asset’s HMM marginal distribution as expected. The SIM, by contrast, produced KS pass rates of 79.5% for NVDA, 92.1% for JNJ, and only 27.5% for JPM, which reflected the inability of a linear factor model to reproduce asset-specific tail structure when the factor loading was moderate ($R^2 = 0.48$ for JPM). This gap demonstrated that the multi-asset quality limitation identified in the 424-asset SIM analysis was attributable to the dependence specification, not to the underlying HMM marginal models. While copulas preserved marginals, the key question was whether they also captured cross-asset dependence. We measured this using the Frobenius norm of the correlation error and the mean absolute error across all six pairwise correlations (Figure 5, panel b). The Student-t copula achieved the best correlation reproduction (Frobenius error 0.134, SE 0.001; pairwise MAE 0.039), followed by the Gaussian copula (0.168; 0.055) and the vine copula (0.287; 0.099). The vine copula’s poor correlation reproduction in this four-asset setting was attributable to overfitting: with only six bivariate edges and five candidate families per edge, the AIC-selected model captured local pairwise structure at the expense of global correlation coherence. The Student-t copula’s advantage over the Gaussian copula arose from its ability to model tail dependence; at the fitted $\nu = 5$ degrees of freedom, extreme co-movements across assets occurred substantially more frequently than a Gaussian dependence structure would produce, which better matched the observed concentration of joint drawdowns in the historical data.

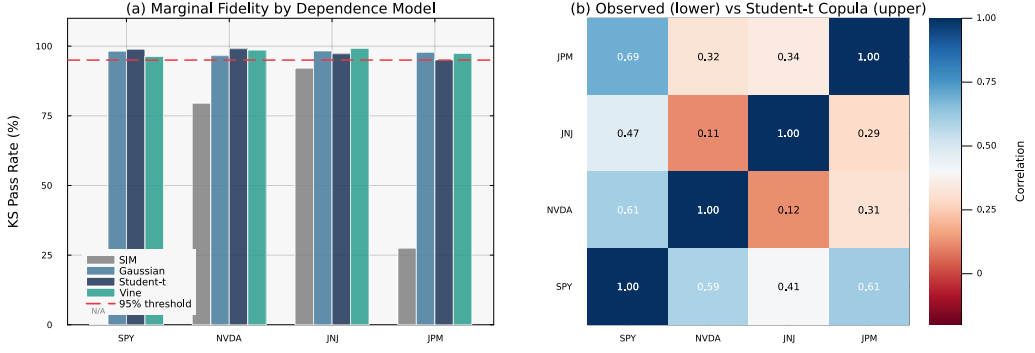


Fig. 5. Dependence model comparison for the four-asset portfolio (SPY, NVDA, JNJ, JPM). Panel (a): KS pass rates by ticker and dependence model. All three copula specifications preserve per-asset marginal fidelity above 95%, while the SIM degrades to 27.5% for JPM due to low factor loading ($R^2 = 0.48$). The dashed line marks the 95% threshold. Panel (b): observed pairwise correlations (lower triangle) vs. Student-t copula simulated correlations (upper triangle). The Student-t copula reproduces the cross-asset dependence structure with a pairwise MAE of 0.039; the largest discrepancy is SPY-JNJ (observed 0.47, simulated 0.41).

The vine copula's underperformance had a revealing explanation: the AIC-based selection procedure chose the Student-t bivariate copula at all six edges across all three vine tree levels ($\Delta\text{AIC} > 60$ over every alternative [13]), with no Clayton, Gumbel, or Frank families selected. This was unexpected given that the financial risk literature frequently posits asymmetric tail dependence in equity returns [3, 42]. The fitted degrees-of-freedom parameter varied systematically: $\nu = 4-5$ for direct market pairs (SPY-NVDA, SPY-JNJ, SPY-JPM), indicating heavy symmetric tail dependence, and $\nu = 10-30$ for conditional pairs after removing the market factor, indicating that most tail dependence was attributable to the common market component. Because every vine edge selected the same family, the vine's flexibility provided no advantage over a single Student-t copula while adding estimation noise from the sequential h-function transformations. For these four U.S. large-cap equities over this sample period, a single Student-t copula was a parsimonious and sufficient specification; whether this symmetry persists across broader asset universes or crisis periods remains an open question.

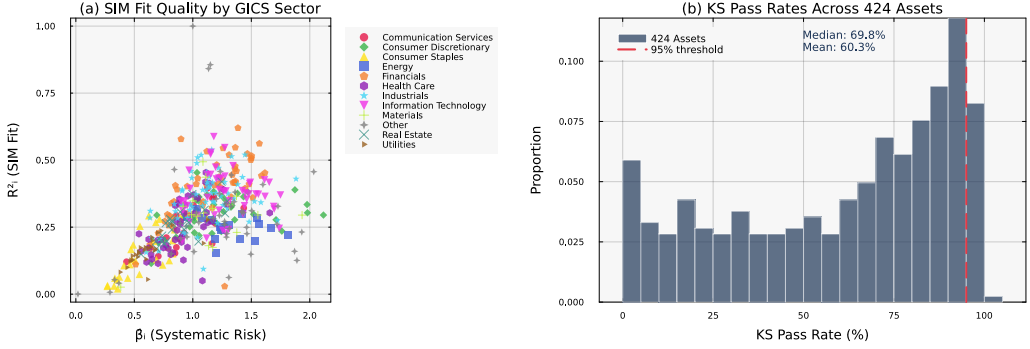


Fig. 6. Multi-asset extension via the Single-Index Model (SIM) across 424 S&P 500 constituents. Panel (a) shows the SIM regression fit quality (R^2 vs. $\hat{\beta}_i$) by GICS sector; assets with higher systematic risk exposure exhibit stronger factor model fit. Panel (b) shows the distribution of KS pass rates across the full asset universe (median 66.7%, mean 58.4%), confirming that the SIM captures most of the distributional structure for high- β assets but degrades for low-correlation names.

5 Discussion

In this study we developed a hybrid hidden Markov model that combines a fine-grained quantile-based state partition ($N = 100$) with a Poisson jump-duration mechanism to generate synthetic equity excess growth rate paths. We benchmarked it against seven alternative generators on SPY using a multi-dimensional quality evaluation (KS/AD pass rates, Wasserstein-1 and Hellinger distances, kurtosis matching, ACF-MAE, and quantile coverage) and extended it to a 424-asset universe via both a Single-Index Model factor decomposition and copula-based dependence specifications. The results showed that no single model dominated all quality dimensions: GARCH captured temporal structure but failed distributional tests, the GRU learned dynamics but suffered variance collapse, and the standard HMM matched distributions but could not reproduce volatility clustering. HMM-WJ was the only model to improve meaningfully on both dimensions, occupying the Pareto frontier between distributional fidelity and temporal structure. The paragraphs below interpret why the jump mechanism produces this advantage, discuss the practical implications for synthetic data generation and risk management, and identify the limitations that bound these conclusions.

The comparison between HMM-NJ and HMM-WJ isolated the contribution of the Poisson jump-duration mechanism. Without jumps, the Markov chain exited tail states within one to two steps on average, producing an ACF of $|G_t|$ that decayed too quickly. Introducing jumps forced the model to dwell in high-volatility regimes for empirically realistic durations [12, 49]. The HSMM baseline provided a direct test of the alternative semi-Markov approach: despite using the same Student-t emissions, the HSMM's coarse $K = 8$ state partition caused empirical dwell times to average only 1.1–1.8 steps, collapsing to memoryless behavior (82% KS vs. 98% for HMM-WJ; ACF-MAE at the i.i.d. floor). This advantage was independent of the emission distribution choice (Table T4 in Online Appendix S9). Our framework resolves this resolution-persistence tradeoff by maintaining fine state resolution ($N = 100$) for distributional fidelity while enforcing tail-state persistence externally via the jump mechanism. This tradeoff shaped the broader model comparison: no single model dominated all quality dimensions simultaneously. HMM-WJ occupied the Pareto frontier between distributional fidelity and temporal structure: it reduced ACF-MAE by 28% relative to the i.i.d. floor while maintaining pass rates above 91% of the Bootstrap ceiling. GARCH achieved the best temporal fidelity but catastrophic distributional failure (5.5% KS); the

GRU learned temporal dynamics effectively but suffered variance collapse (0.6% KS). The value of HMM-WJ lay in avoiding each alternative’s severe failure mode. Crucially, the volatility-clustering strength is continuously tunable via ϵ , and the grid search selected the value that jointly optimized temporal and distributional quality.

These tradeoffs carry practical implications. Generative models that reproduce the stylized facts of financial markets enable stress testing under market regimes that are statistically plausible but not historically observed, addressing a key limitation of scenario libraries derived solely from empirical records. The interpretable regime structure, where each hidden state corresponds to a quantile-defined segment of the excess growth rate distribution, facilitates communication between quantitative analysts and risk managers; states can be labeled (e.g., *crash*, *bear*, *neutral*, *bull*, *rally*) and linked to economic narratives. The model’s computational efficiency further supports Monte Carlo-based risk metrics such as Value-at-Risk and Conditional Value-at-Risk at scale across large asset universes [29]. This computational efficiency stems from the avoidance of the Baum-Welch EM algorithm, made possible by the quantile-based state partition that explicitly assigns each observation to a hidden state, eliminates convergence sensitivity to initialization and reduces computational cost. This property matters for large-scale pipelines. Fitting the model to 424 assets requires only empirical counting and a two-parameter grid search, making it feasible to regenerate synthetic scenarios on a daily or weekly cadence as new data arrive. For synthetic data applications where privacy is a concern, such as sharing simulated portfolios or stress-test scenarios with external parties, the framework offers implicit privacy protection: synthetic paths are generated from the learned Markov chain and emission distribution rather than slightly modified copies of historical records. While this generative structure reduces the risk of memorization, the framework does not provide formal differential privacy guarantees [21]; quantifying membership inference risk and integrating calibrated noise mechanisms remain directions for future work [53]. Beyond the model itself, the multi-metric evaluation approach used here, combining KS/AD pass rates with Wasserstein/Hellinger distances and ACF-MAE, can be applied to any time series generator where temporal dependence is critical to downstream applications [55].

Several limitations qualify these conclusions. First, the model assumes stationarity of the transition matrix and jump hyperparameters across the full in-sample window. The out-of-sample ACF analysis (panel d) provided direct evidence of this limitation. The IS-calibrated (ϵ^* , λ^*) underestimated volatility clustering persistence in the 2025 test window, consistent with the elevated macro uncertainty of that period. Structural breaks in market microstructure, such as those accompanying regulatory changes or the COVID-19 shock, may not be adequately captured by a single static transition matrix. Relatedly, the quantile-based state partition is fixed at estimation time and does not adapt to evolving volatility regimes, potentially overweighting the central mass relative to the tails during prolonged market stress. Third, the Single-Index Model extension imposes a linear, single-factor decomposition whose median IS KS pass rate of 66.7% across 424 assets (well below the 95% level expected under a correct model) confirmed that it could not fully capture asset-specific tail behavior, skewness, or sector dynamics [25, 51]. The copula comparison demonstrated that replacing the SIM with a Student-t copula restored marginal KS pass rates above 95% for all tested assets while also reproducing pairwise correlations to within 4 percentage points on average (MAE of 0.039 on the $[-1, 1]$ correlation scale); this confirmed that the HMM marginal models were not the bottleneck and that the multi-asset quality gap was attributable to the dependence specification. The architecture is modular, and substituting a copula for the SIM does not require modifying the underlying Markov chain machinery; however, copula estimation scales as $O(d^2)$ in the number of assets, which may limit applicability to very large universes where the SIM’s $O(d)$ scaling remains advantageous. Finally, the out-of-sample evaluation covers a single 249-day window (full calendar

year 2025), which limits the statistical power of out-of-sample inference; a rolling evaluation would provide a more robust assessment of generalization performance.

A separate structural concern is that the jump mechanism produces a bimodal ensemble. At $\epsilon^* = 10^{-4}$, approximately 76% of simulated paths contain no jump events and therefore exhibit no volatility clustering, while the remaining 24% exhibit pronounced clustering (Online Appendix S7). For applications that draw a small number of paths (e.g., a single stress-test scenario), this means the majority of draws will lack temporal structure entirely. Users requiring volatility clustering in every path should either condition on jump-containing paths or increase ϵ at the cost of marginal fidelity. The optimal ϵ^* sits at the lower boundary of the search grid, and extending the grid below 10^{-4} could reveal that even rarer jumps are preferred by the objective; future work should explore whether alternative objective formulations that penalize the fraction of jump-free paths could address this limitation. On the data side, the excess growth rate calculation subtracts a constant risk-free rate r_f derived from STRIPS yields, but the effective Federal Funds rate varied from 0.09% to 5.33% over the 2014–2024 sample period. This time-varying discrepancy introduces a slow-moving bias into the growth rate series that the model may interpret as regime structure rather than interest-rate dynamics. Additionally, the 2014–2024 training window includes the March 2020 COVID-19 crash, which likely dominates the empirical kurtosis estimate and the transition matrix’s tail entries; whether the model’s calibration generalizes to periods without such an extreme event remains untested. Finally, the two-sample KS and AD tests assume i.i.d. observations, but each simulated path exhibits temporal dependence by construction. Volatility clustering induces positive autocorrelation in the tail-indicator process $\mathbf{1}(G_t \leq x)$, reducing the effective sample size below T and potentially inflating pass rates slightly relative to what a block-bootstrap calibration would produce. The Wasserstein-1 and Hellinger distances carry no such sampling-distribution assumption and produce a consistent model ordering, corroborating the fidelity conclusions independently of this caveat.

6 Conclusion

The key finding of this study was that no single generative model dominated all quality dimensions: GARCH(1,1) better reproduced volatility clustering but failed distributional tests, the standard HMM passed distributional tests but could not generate persistent high-volatility regimes, and the hybrid framework, which augmented quantile-defined regime switching with a Poisson jump-duration mechanism, delivered the most balanced performance by achieving high distributional pass rates both in-sample and out-of-sample with partial reproduction of volatility clustering governed by two tunable hyperparameters. Direct frequentist estimation eliminated the computational cost and initialization sensitivity of the EM algorithm, scaling the approach to a 424-asset pipeline, and replacing the Single-Index Model's linear factor structure with a Student-t copula substantially improved per-asset distributional accuracy for assets that did not closely track the index. Several directions for future work follow from these results: time-varying transition matrices, estimated via rolling windows or Bayesian online learning, would allow the model to adapt to structural regime shifts; scaling the Student-t copula to larger asset universes via factor copulas or truncated vine structures would extend the per-asset accuracy gains to full portfolio applications; and embedding the hybrid HMM within a portfolio optimization loop, where synthetic scenarios drive tail-risk objectives, would provide an end-to-end test of the framework's practical value.

Conflict of Interest Statement

The authors declare that the research was conducted without any commercial or financial relationships that could potentially create a conflict of interest.

Author Contributions

J.V. directed the study. A.A. developed the model and simulation code, conducted the in-sample and out-of-sample analysis, and generated the figures. Both authors edited and reviewed the final manuscript.

Data Availability Statement

The simulation scripts, training and testing data, and all code to reproduce the results in this paper are available under a Massachusetts Institute of Technology (MIT) license from the paper repository: <https://github.com/varnerlab/HMM-w-jumps-paper.git>. The core model logic (fitting, simulation, decoding, and validation) is implemented in the JumpHMM.jl package: <https://github.com/varnerlab/JumpHMM.jl.git>.

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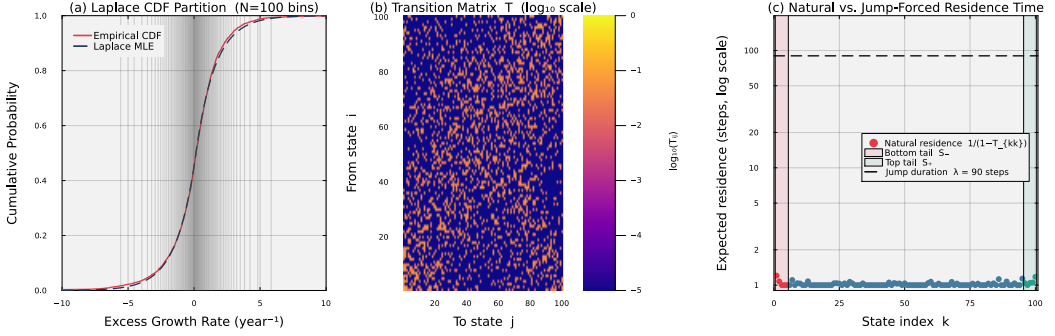


Fig. S1. Fitted model internals for SPY with $N = 100$ states. Panel (a) overlays the fitted Laplace CDF on the empirical CDF; the 99 vertical lines mark the equal-probability quantile boundaries that define the N hidden states. The close agreement confirms that the Laplace distribution is an accurate marginal model for the excess growth rate data. Panel (b) displays the empirical transition matrix T in $\log_{10}$ color scale; the dominant near-diagonal band reflects short-range regime persistence. Panel (c) plots the expected natural residence time $1/(1 - T_{kk})$ for each state on a $\log_{10}$ scale, with the bottom tail set S_- (states 1–5, red shading) and top tail set S_+ (states 96–100, teal shading) highlighted. Under pure Markovian dynamics every state, including the tail states, exits within 1–2 steps on average, far too quickly to reproduce empirical volatility clustering. The dashed horizontal line marks the mean mean jump duration $\lambda^* = 100$ steps at the grid-search optimum, quantifying the two-order-of-magnitude gap between natural residence times and the jump-duration mechanism.

Online Appendix

S1. Fitted model internals

This section visualizes the internal components of the fitted HMM-WJ model for SPY with $N = 100$ states. We constructed the model by fitting a Laplace distribution to the observed excess growth rate series (2014–2024, 2,766 trading days), partitioning the support into N equal-probability quantile bins, and estimating the 100×100 transition matrix from consecutive state assignments (Algorithm 2). Figure S1 displays three diagnostics. Panel (a) confirms that the Laplace marginal closely matches the empirical CDF, validating the distributional assumption underlying the state partition. Panel (b) reveals the banded structure of the transition matrix: most probability mass concentrates near the diagonal, reflecting the tendency of markets to transition between adjacent states rather than making large jumps. Panel (c) exposes the key limitation of the pure Markov dynamics: the natural residence time $1/(1 - T_{kk})$ for every state, including the extreme tails, is only 1–2 steps. This is far too short to reproduce the persistent volatility clustering observed in equity returns, motivating the Poisson jump-duration mechanism described in Section 3.3 of the main text.

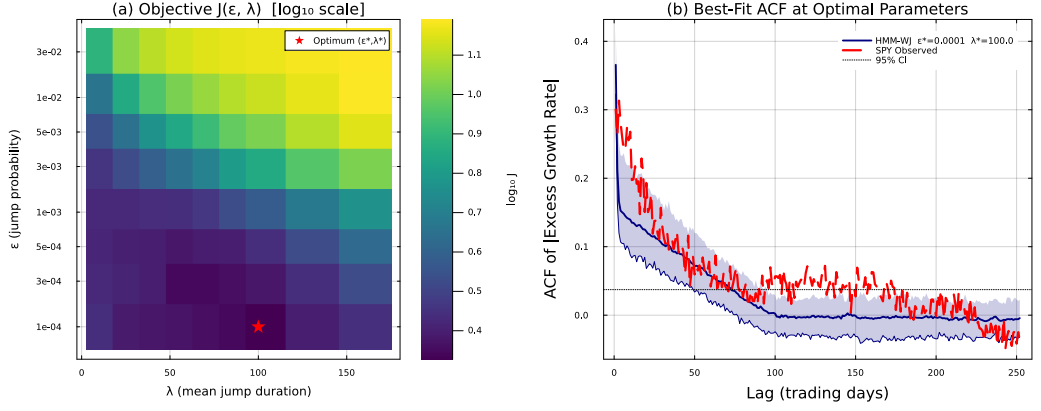


Fig. S2. Hyperparameter grid search over (ϵ, λ) for SPY ($N = 100, 200$ paths per grid point). Panel (a): heatmap of the objective function $J(\epsilon, \lambda)$ in $\log_{10}$ scale over the full search grid ($\epsilon \in [10^{-4}, 2.5 \times 10^{-2}]$, $\lambda \in [10, 160]$); the red star marks the optimal pair ($\epsilon^* = 10^{-4}$, $\lambda^* = 100$). The clear minimum confirms that the jump hyperparameters are jointly identifiable from data; ϵ^* sits at the lower boundary of the search grid, indicating that smaller tail-entry probabilities are consistently preferred. Panel (b): ACF of $|G_t|$ at the optimal parameters computed from 500 jump-containing paths (10th–90th percentile band shaded; mean curve solid navy) compared with the SPY observed ACF (dashed red) and the 95% significance band (dotted black).

S2. Hyperparameter grid search landscape

The jump mechanism introduces two hyperparameters: the tail-entry probability ϵ and the mean jump duration λ . We selected these via the multi-objective grid search described in Algorithm 4, sweeping $\epsilon \in [10^{-4}, 2.5 \times 10^{-2}]$ (eight points) and $\lambda \in [10, 160]$ (nine points) with 200 simulated paths per grid point. The objective $J(\epsilon, \lambda)$ balances two terms: the squared error between observed and simulated absolute-return ACFs (lags 1–252) and a kurtosis penalty weighted by $w_K = 0.20$. Figure S2 presents the results. Panel (a) shows that J is well-behaved over the grid, with a clear minimum confirming that the two hyperparameters are jointly identifiable from data. The optimal values ($\epsilon^* = 10^{-4}$, $\lambda^* = 100$) indicate that the data favor rare but long-lived tail excursions; ϵ^* falls at the lower boundary of the ϵ grid, while λ^* is an interior optimum. Panel (b) validates that the ACF of $|G_t|$ at these optimal parameters closely tracks the observed SPY autocorrelation structure, with the simulated 10th–90th percentile envelope (computed from 500 paths) enveloping the empirical curve over the full 252-lag horizon.

Table T1. Sensitivity of model performance to state resolution N . All metrics computed over 1,000 simulated in-sample paths at $\alpha = 0.05$. ACF-MAE = mean absolute error of the ACF of $|G_t|$ over lags 1–252. Standard errors in parentheses: binomial SE for pass rates, bootstrap SE ($B = 500$) for ACF-MAE. Values at $N = 100$ differ slightly from Table 2 because the grid search was re-run independently for each N , producing a different (ϵ^*, λ^*) optimum due to Monte Carlo variability.

Model	N	KS pass (%)	AD pass (%)	ACF-MAE
HMM-NJ	200	99.6 (0.2)	98.5 (0.4)	0.059
HMM-NJ	150	99.6 (0.2)	98.4 (0.4)	0.059
HMM-NJ	100	99.0 (0.3)	98.8 (0.3)	0.059
HMM-NJ	90	99.4 (0.2)	98.1 (0.4)	0.059
HMM-NJ	60	99.4 (0.2)	98.1 (0.4)	0.059
HMM-NJ	30	98.9 (0.3)	97.9 (0.5)	0.059
HMM-WJ	200	96.0 (0.6)	88.6 (1.0)	0.049
HMM-WJ	150	96.9 (0.5)	88.8 (1.0)	0.050
HMM-WJ	100	96.8 (0.6)	91.7 (0.9)	0.053
HMM-WJ	90	97.4 (0.5)	91.0 (0.9)	0.052
HMM-WJ	60	96.8 (0.6)	92.4 (0.8)	0.054
HMM-WJ	30	98.0 (0.4)	96.8 (0.6)	0.057

S3. State resolution sensitivity

A natural concern is whether the model’s performance depends critically on the choice of N , the number of discrete states. To investigate this, we re-estimated the full pipeline (Laplace fit, quantile partition, transition matrix, and, for HMM-WJ, the grid search over ϵ and λ) for $N \in \{30, 60, 90, 100, 150, 200\}$ and evaluated each configuration on 1,000 simulated in-sample paths. Table T1 reports KS and AD pass rates at the $\alpha = 0.05$ level along with ACF-MAE (mean absolute error of the absolute-return ACF over lags 1–252). Standard errors are computed via the binomial formula for pass rates and bootstrap resampling ($B = 500$) for ACF-MAE.

For HMM-NJ, distributional fidelity is uniformly high across all resolutions (KS $\geq 98.9\%$, AD $\geq 97.9\%$), confirming that the Laplace quantile partition produces well-calibrated marginals regardless of N . ACF-MAE is effectively constant at 0.059, reflecting the absence of any volatility-clustering mechanism. For HMM-WJ, KS pass rates remain above 96% across all N , while AD pass rates range from 88.6% ($N = 200$) to 96.8% ($N = 30$). The slight decline at high N is expected: finer partitions produce more sparsely populated tail states, making the Anderson-Darling test (which weights tail discrepancies heavily) more sensitive. Crucially, ACF-MAE improves monotonically from 0.057 ($N = 30$) to 0.049 ($N = 200$), demonstrating that the jump mechanism’s ability to reproduce volatility clustering strengthens with finer state resolution. The operating point $N = 100$ balances these two effects, achieving strong performance on all three metrics simultaneously.

Algorithm 1 Hybrid jump-diffusion simulation with persistent volatility

Require: Model parameters $\mathbf{T}, \bar{\pi}, \epsilon, \lambda, N_{tail}, p_{neg}$, Total Steps M

Ensure: Simulated state sequence $S_{1:M}$

```

1: Define tail-states  $\mathcal{S}_{bottom} = \{1, \dots, N_{tail}\}$  and  $\mathcal{S}_{top} = \{N - N_{tail} + 1, \dots, N\}$ .
2: Sample initial state  $S_1 \sim \text{Categorical}(\bar{\pi})$ .
3: Set  $counter \leftarrow 2$ .
4: while  $counter \leq M$  do
5:   Sample  $u \sim \text{Uniform}(0, 1)$ .
6:   if  $u < \epsilon$  then
7:     Sample jump-duration  $K \sim \text{Poisson}(\lambda)$ .
8:     for  $j = 1$  to  $K$  while  $counter \leq M$  do
9:       Sample  $w \sim \text{Uniform}(0, 1)$ .
10:      if  $w < p_{neg}$  then
11:         $S_{counter} \sim \text{Uniform}(\mathcal{S}_{bottom})$  {Bias toward negative-tail events}
12:      else
13:         $S_{counter} \sim \text{Uniform}(\mathcal{S}_{top})$ 
14:      end if
15:       $counter \leftarrow counter + 1$ .
16:    end for
17:  else
18:    Sample  $S_{counter} \sim \text{Categorical}(\mathbf{T}_{S_{counter-1}, \cdot})$ .
19:     $counter \leftarrow counter + 1$ .
20:  end if
21: end while
22: return  $S_{1:M}$ 

```

S4. Algorithms

The four algorithms below detail the complete computational pipeline. Algorithm 1 generates synthetic state sequences from the fitted HMM with jump-duration dynamics (referenced in the main text). The remaining three algorithms detail model construction, state decoding, and hyperparameter optimization.

Algorithm 2 describes model construction: fitting a Laplace distribution to the observed excess growth rates, partitioning the support into N equal-probability bins, assigning each observation to a discrete state, and estimating the transition matrix by row-normalizing the state-to-state count matrix. Algorithm 3 converts a simulated discrete state sequence back into continuous excess growth rates by sampling from a location-scale Student-t distribution ($\nu = 5$) within each state, parameterized by the state-conditional sample mean and standard deviation. Algorithm 4 performs the multi-objective grid search over the jump hyperparameters (ϵ, λ), minimizing a composite objective that penalizes both ACF mismatch and kurtosis deviation relative to the observed series.

S5. Cross-asset generalization

To assess whether the framework generalized beyond the SPY market factor, we fitted standalone HMM-NJ and HMM-WJ models to three individual equities spanning distinct risk profiles: NVDA (high-beta technology), JNJ (low-beta health care), and JPM (moderate-beta financials). Each ticker was fitted independently using the same $N = 100$ Laplace quantile partition and the same grid search procedure described in Section 3.3 of the main text, with 1,000 simulated paths evaluated against the observed series.

Table T2. Cross-asset generalization of HMM-NJ and HMM-WJ to individual equities ($N = 100, 1,000$ simulated paths, $\alpha = 0.05$). Each ticker was fitted independently with a per-asset grid search over (ϵ, λ) . Standard errors in parentheses: binomial SE for pass rates, $\text{std}/\sqrt{n}$ for kurtosis, bootstrap SE ($B = 500$) for ACF-MAE.

Ticker	Model	KS pass (%)	AD pass (%)	Excess kurtosis	ACF-MAE
<i>In-sample (2,766 trading days, 2014–2024)</i>					
				Observed: 5.0	
NVDA	HMM-NJ	99.3 (0.3)	99.1 (0.3)	4.4 (0.02)	0.043 (<0.001)
NVDA	HMM-WJ	91.7 (0.9)	77.5 (1.3)	4.1 (0.03)	0.028 (0.001)
				Observed: 8.7	
JNJ	HMM-NJ	99.8 (0.1)	99.2 (0.3)	6.8 (0.04)	0.027 (<0.001)
JNJ	HMM-WJ	99.8 (0.1)	99.1 (0.3)	6.7 (0.04)	0.026 (<0.001)
				Observed: 8.0	
JPM	HMM-NJ	99.6 (0.2)	99.3 (0.3)	6.7 (0.03)	0.045 (<0.001)
JPM	HMM-WJ	92.8 (0.8)	79.6 (1.3)	6.3 (0.03)	0.032 (0.001)
<i>Out-of-sample (249 trading days, 2025)</i>					
				Observed: 7.0	
NVDA	HMM-NJ	99.0 (0.3)	97.7 (0.5)	3.8 (0.07)	0.039 (<0.001)
NVDA	HMM-WJ	97.1 (0.5)	95.6 (0.6)	3.7 (0.07)	0.037 (<0.001)
				Observed: 6.4	
JNJ	HMM-NJ	73.2 (1.4)	59.4 (1.6)	5.5 (0.11)	0.034 (<0.001)
JNJ	HMM-WJ	70.3 (1.4)	59.7 (1.6)	5.6 (0.11)	0.034 (<0.001)
				Observed: 6.7	
JPM	HMM-NJ	77.6 (1.3)	78.1 (1.3)	5.9 (0.09)	0.032 (<0.001)
JPM	HMM-WJ	76.9 (1.3)	76.4 (1.3)	5.8 (0.09)	0.031 (<0.001)

Table T2 reports the results. In-sample KS pass rates exceeded 91% for all three tickers under both HMM-NJ and HMM-WJ, confirming that the Student-t emission structure captured the marginal distribution of individual equities with diverse tail characteristics. The grid search selected $\epsilon^* = 10^{-4}$ for all three assets, with $\lambda^* = 160$ for NVDA and JPM (long-lived volatility episodes) and $\lambda^* = 30$ for JNJ (shorter episodes consistent with its weaker volatility clustering). HMM-WJ reduced ACF-MAE relative to HMM-NJ for NVDA (0.043→0.028) and JPM (0.045→0.032), while JNJ showed minimal improvement (0.027→0.026), consistent with its lower baseline autocorrelation in absolute returns.

Out-of-sample performance varied across tickers. NVDA maintained strong distributional fidelity (KS 97.1%, AD 95.6%), while JNJ and JPM showed more pronounced OoS degradation (KS 70–77%), reflecting asset-specific regime shifts in 2025 that the stationary IS-calibrated parameters could not fully capture. These results confirmed that the HMM-WJ framework was not specific to ETFs or broad market indices; it generalized to individual equities, with the grid search automatically adapting the jump parameters to each asset’s volatility dynamics.

Algorithm 2 Empirical hidden Markov model construction via Laplace partitioning

Require: Price history $P_{1:T}$, Risk-free rate r_f , Number of states N

Ensure: Transition matrix $\mathbf{T}$, Quantile boundaries $\mathbf{Q}$

- 1: Compute excess growth rates $R_t = \frac{1}{\Delta t} \ln(P_t/P_{t-1}) - r_f$ for all t .
 - 2: Fit Laplace parameters (μ_L, b_L) to the series $\mathbf{R}$ using maximum-likelihood estimation.
 - 3: Define interior boundaries $Q_k = F_L^{-1}(k/N; \mu_L, b_L)$ for $k \in \{1, \dots, N-1\}$, where F_L^{-1} is the inverse CDF of the Laplace distribution.
 - 4: Set finite outer bounds $Q_0 = F_L^{-1}(0.001; \mu_L, b_L)$ and $Q_N = F_L^{-1}(0.999; \mu_L, b_L)$.
 - 5: Assign each excess growth rate R_t to a discrete state $s_t \in \{1, \dots, N\}$ based on the boundaries in $\mathbf{Q} = \{Q_k\}$.
 - 6: Initialize count matrix $\mathbf{C} \in \mathbb{R}^{N \times N}$ and transition matrix $\mathbf{T} \in \mathbb{R}^{N \times N}$ with zeros.
 - 7: **for** $t = 1$ to $T - 1$ **do**
 - 8: Identify transition $i = s_t$ to $j = s_{t+1}$.
 - 9: $\mathbf{C}_{i,j} \leftarrow \mathbf{C}_{i,j} + 1$.
 - 10: **end for**
 - 11: **for** $i = 1$ to N **do**
 - 12: $\mathbf{T}_{i,:} \leftarrow \mathbf{C}_{i,:} / \sum_{j=1}^N \mathbf{C}_{i,j}$ {Row-wise normalization}
 - 13: **end for**
 - 14: **return** $\mathbf{T}, \mathbf{Q}$
-

Algorithm 3 State decoding and continuous excess growth rate reconstruction

Require: Simulated state sequence $S_{1:M}$, Encoded training observations, Degrees of freedom $\nu = 5$

Ensure: Reconstructed continuous excess growth rates $\hat{R}_{1:M}$

```
1: for  $k = 1$  to  $N$  do
2:   Set  $\mu_k = \text{mean}(\{G_t : S_t^{\text{train}} = k\})$ .
3:   Set  $\sigma_k = \text{std}(\{G_t : S_t^{\text{train}} = k\})$ .
4: end for
5: Initialize the reconstructed excess growth rate vector  $\hat{\mathbf{G}}$  of length  $M$ .
6: for  $t = 1$  to  $M$  do
7:   Retrieve the simulated state  $k = S_t$ .
8:   Sample  $\hat{G}_t = \mu_k + \sigma_k \cdot Z$ , where  $Z \sim t_\nu$ .
9: end for
10: return  $\hat{\mathbf{G}}_{1:M}$ 
```

Algorithm 4 Multi-objective grid search for jump hyperparameters

Require: Observed growth rates $\mathbf{R}_{obs}$, ACF max lag $L = 252$, Number of paths $P = 200$, Time steps $M = 2766$

Require: Grid: $\epsilon \in \{10^{-4}, 2.5 \times 10^{-4}, 5 \times 10^{-4}, 10^{-3}, 2.5 \times 10^{-3}, 5 \times 10^{-3}, 10^{-2}, 2.5 \times 10^{-2}\}$, $\lambda \in \{10, 25, 40, 55, 70, 85, 100, 130, 160\}$

Require: Kurtosis penalty weight $w_K = 0.20$

Ensure: Optimal jump hyperparameters (ϵ^*, λ^*)

- 1: Calculate observed absolute ACF: $A_{obs}(\tau) = \text{ACF}(|\mathbf{R}_{obs}|, \tau)$ for $\tau \in \{1, \dots, L\}$.
 - 2: Calculate observed global kurtosis: $K_{obs} = \text{Kurtosis}(\mathbf{R}_{obs})$.
 - 3: Initialize minimum error $E_{min} \leftarrow \infty$.
 - 4: **for** each ϵ in grid space **do**
 - 5: **for** each λ in grid space **do**
 - 6: Initialize simulated ACF accumulator $\mathbf{A}_{sum} \leftarrow \mathbf{0}$ and kurtosis accumulator $K_{sum} \leftarrow 0$.
 - 7: **for** $p = 1$ to P **do**
 - 8: Simulate path using Algorithm 1 to obtain state sequence $S_{1:M}$.
 - 9: Decode sequence using Algorithm 3 to obtain continuous rates $\hat{\mathbf{R}}^{(p)}$.
 - 10: $\mathbf{A}_{sum}(\tau) \leftarrow \mathbf{A}_{sum}(\tau) + \text{ACF}(|\hat{\mathbf{R}}^{(p)}|, \tau)$ for all τ .
 - 11: $K_{sum} \leftarrow K_{sum} + \text{Kurtosis}(\hat{\mathbf{R}}^{(p)})$.
 - 12: **end for**
 - 13: Compute ensemble averages: $\bar{A}_{sim}(\tau) = \mathbf{A}_{sum}(\tau)/P$ and $\bar{K}_{sim} = K_{sum}/P$.
 - 14: Compute objective: $J = \sum_{\tau=1}^L \left(A_{obs}(\tau) - \bar{A}_{sim}(\tau) \right)^2 + w_K \left(K_{obs} - \bar{K}_{sim} \right)^2$.
 - 15: **if** $J < E_{min}$ **then**
 - 16: $E_{min} \leftarrow J$
 - 17: $(\epsilon^*, \lambda^*) \leftarrow (\epsilon, \lambda)$
 - 18: **end if**
 - 19: **end for**
 - 20: **end for**
 - 21: **return** ϵ^*, λ^*
-

S6. Copula dependence models: mathematical details

This section provides the mathematical foundations for the copula-based dependence models introduced in Section 3. The main text describes these models at a conceptual level; here we give the densities, fitting procedures, and simulation algorithms in full.

Sklar's theorem and the copula framework. Let $\mathbf{G} = (G^{(1)}, \dots, G^{(d)})$ denote a d -dimensional random vector of excess growth rates with joint CDF H and marginal CDFs $F_1, \dots, F_d$. Sklar's theorem [22] states that there exists a copula $C : [0, 1]^d \rightarrow [0, 1]$ such that

$$H(g_1, \dots, g_d) = C(F_1(g_1), \dots, F_d(g_d)). \quad (\text{S1})$$

If the marginals are continuous, C is unique. The copula C captures all dependence structure independently of the marginals, which is why it pairs naturally with per-asset HMM marginal models: each asset's marginal distribution is determined by its fitted HMM, while the copula governs how extreme events co-occur across assets.

Probability integral transform (PIT). To fit a copula, we first transform the observed growth rates to uniform marginals. Given n observations $g_{1,t}, \dots, g_{d,t}$ for $t = 1, \dots, n$, we compute pseudo-uniform observations

$$u_{j,t} = \frac{\text{rank}(g_{j,t})}{n+1}, \quad j = 1, \dots, d, \quad (\text{S2})$$

where $\text{rank}(g_{j,t})$ is the ordinal rank of $g_{j,t}$ among $\{g_{j,1}, \dots, g_{j,n}\}$. The denominator $n+1$ (rather than n) ensures that $u_{j,t} \in (0, 1)$, avoiding boundary singularities in the copula density. This nonparametric PIT avoids misspecification of the marginals.

Gaussian copula. The Gaussian copula with correlation matrix Σ has CDF

$$C_{\text{Ga}}(\mathbf{u}; \Sigma) = \Phi_d(\Phi^{-1}(u_1), \dots, \Phi^{-1}(u_d); \Sigma), \quad (\text{S3})$$

where Φ is the standard normal CDF and $\Phi_d(\cdot; \Sigma)$ is the d -dimensional normal CDF with correlation matrix Σ . For a bivariate pair (u, v) with correlation ρ , the copula density is

$$c_{\text{Ga}}(u, v; \rho) = \frac{1}{\sqrt{1-\rho^2}} \exp\left(-\frac{\rho^2(x^2 + y^2) - 2\rho xy}{2(1-\rho^2)}\right), \quad (\text{S4})$$

where $x = \Phi^{-1}(u)$ and $y = \Phi^{-1}(v)$. The parameter ρ is estimated from Kendall's τ via $\rho = \sin(\pi\tau/2)$, which is the exact inversion of the relationship $\tau = (2/\pi) \arcsin(\rho)$ for the Gaussian copula. This rank-based estimator is robust to outliers and does not depend on the marginal distributions; it generally differs from both the Pearson correlation of the raw growth rates and the Pearson correlation of the PIT-transformed uniforms. The Gaussian copula has zero upper and lower tail dependence coefficients, $\lambda_U = \lambda_L = 0$, regardless of ρ [22]; in other words, the probability of jointly extreme observations vanishes faster than linearly in the tail.

Student-t copula. The Student-t copula with correlation matrix Σ and ν degrees of freedom has CDF

$$C_t(\mathbf{u}; \Sigma, \nu) = t_{d,\nu}(t_v^{-1}(u_1), \dots, t_v^{-1}(u_d); \Sigma), \quad (\text{S5})$$

where t_ν is the univariate Student-t CDF and $t_{d,\nu}(\cdot; \Sigma)$ is the d -variate Student-t CDF with ν degrees of freedom and correlation Σ . For a bivariate pair the log-density is

$$\ln c_t(u, v; \rho, \nu) = \ln \Gamma_{\frac{\nu+2}{2}} + \ln \Gamma_{\frac{\nu}{2}} - 2 \ln \Gamma_{\frac{\nu+1}{2}} - \frac{1}{2} \ln(1-\rho^2) + T_3 + T_4, \quad (\text{S6})$$

where $x = t_v^{-1}(u)$, $y = t_v^{-1}(v)$, and

$$T_3 = -\frac{\nu+2}{2} \ln \left(1 + \frac{x^2 + y^2 - 2\rho xy}{\nu(1-\rho^2)} \right), \quad (S7)$$

$$T_4 = \frac{\nu+1}{2} \left(\ln \left(1 + \frac{x^2}{\nu} \right) + \ln \left(1 + \frac{y^2}{\nu} \right) \right). \quad (S8)$$

The correlation ρ is estimated from Kendall's τ (as for the Gaussian copula), and the degrees-of-freedom ν is selected by profile maximum likelihood over a discrete grid $\nu \in \{2.5, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$. The Student-t copula has *symmetric* tail dependence: $\lambda_U = \lambda_L > 0$ for all finite ν , with the coefficient given by

$$\lambda_U = \lambda_L = 2 t_{\nu+1} \left(-\sqrt{\frac{(\nu+1)(1-\rho)}{1+\rho}} \right), \quad (S9)$$

which increases as ν decreases (heavier tails). For example, at $\rho = 0.5$ and $\nu = 4$, the tail dependence coefficient is approximately 0.18; this means that, conditional on one asset being in its 1st percentile, there is an 18% probability that the other asset is simultaneously in its 1st percentile, far exceeding what a Gaussian copula would predict (exactly 0%).

C-vine copula decomposition. A d -dimensional copula density can be decomposed into a product of $d(d-1)/2$ bivariate copula densities arranged in a vine structure [1]. The canonical vine (C-vine) factorization is

$$c(\mathbf{u}) = \prod_{k=1}^{d-1} \prod_{j=k+1}^d c_{k,j|1,\dots,k-1}(u_{k|1,\dots,k-1}, u_{j|1,\dots,k-1}), \quad (S10)$$

where $c_{k,j|1,\dots,k-1}$ is a bivariate copula density for the pair (k, j) conditioned on variables $\{1, \dots, k-1\}$, and $u_{k|1,\dots,k-1}$ denotes the conditional CDF (pseudo-observation) obtained by iteratively applying the h -function. The h -function for a bivariate copula C_{ab} is defined as

$$h(u | v; \theta) = \frac{\partial C_{ab}(u, v; \theta)}{\partial v}, \quad (S11)$$

and represents the conditional CDF of U given $V = v$ under the bivariate copula C_{ab} . For tree level k , pseudo-observations are computed as $u_{j|1,\dots,k} = h(u_{j|1,\dots,k-1} | u_{k|1,\dots,k-1}; \hat{\theta}_{k,j})$, where $\hat{\theta}_{k,j}$ is the parameter fitted at that edge.

At each edge, five candidate bivariate families are fitted and the one with the lowest Akaike information criterion (AIC) $= 2p - 2 \ln \hat{L}$ is selected, where p is the number of parameters ($p = 2$ for Student-t, $p = 1$ for all others) and $\hat{L}$ is the maximized likelihood on the pseudo-observations at that edge. The five families and their densities are as follows.

- *Gaussian bivariate copula* (Eq. S4): parameter ρ , no tail dependence.
- *Student-t bivariate copula* (Eq. S6): parameters (ρ, ν) , symmetric tail dependence.
- *Clayton copula*: density $c_{Cl}(u, v; \theta) = (1+\theta)(uv)^{-\theta-1}(u^{-\theta}+v^{-\theta}-1)^{-2-1/\theta}$ for $\theta > 0$, with lower tail dependence $\lambda_L = 2^{-1/\theta}$ and $\lambda_U = 0$. Fitted via $\theta = 2\tau/(1-\tau)$.
- *Gumbel copula*: density involves $A = ((-\ln u)^\theta + (-\ln v)^\theta)^{1/\theta}$ with upper tail dependence $\lambda_U = 2 - 2^{1/\theta}$ and $\lambda_L = 0$. Fitted via $\theta = 1/(1-\tau)$ for $\theta \geq 1$.
- *Frank copula*: density $c_{Fr}(u, v; \theta) = -\theta e^{-\theta(u+v)}(e^{-\theta}-1)/D^2$ where $D = e^{-\theta}-1+(e^{-\theta u}-1)(e^{-\theta v}-1)$, with $\lambda_U = \lambda_L = 0$ (no tail dependence). The parameter θ is obtained by numerically inverting the Debye function relationship $\tau = 1 - 4(\mathcal{D}_1(\theta)-1)/\theta$, where $\mathcal{D}_1(\theta) = \theta^{-1} \int_0^\theta t/(e^t-1) dt$.

Variable ordering. The C-vine requires specifying the root variable at each tree level. We order variables by decreasing sum of absolute Kendall's τ : $\text{order}_k = \arg \max_j \sum_{i \neq j} |\hat{\tau}_{ij}|$ among variables not yet placed. For equity portfolios, this typically places the market index (e.g., SPY) at the root, reflecting its role as the dominant source of co-movement.

Simulation via rank reordering. In the PortfolioModel framework, copula simulation proceeds by rank reordering [22]:

- (1) Simulate n_{paths} independent paths from each asset's marginal HMM, producing a matrix $\mathbf{G}_j^{(\text{ind})} \in \mathbb{R}^{T \times n_{\text{paths}}}$ for $j = 1, \dots, d$.
- (2) Sample $T \times d$ uniform variates $\mathbf{U} \sim C$ from the fitted copula.
- (3) For each asset j and each path i , reorder the simulated observations $G_{j,1:T}^{(i)}$ so that their ranks match the copula-sampled ranks $\text{rank}(U_{j,1:T}^{(i)})$.

This procedure preserves each asset's marginal distribution exactly (since reordering does not change the set of values, only their temporal arrangement) while injecting the cross-asset dependence structure specified by the copula. For the Student-t copula, the uniform variates in step 2 are obtained by sampling from a multivariate Student-t distribution with correlation $\hat{\Sigma}$ and $\hat{\nu}$ degrees of freedom, then applying the univariate Student-t CDF to each margin.

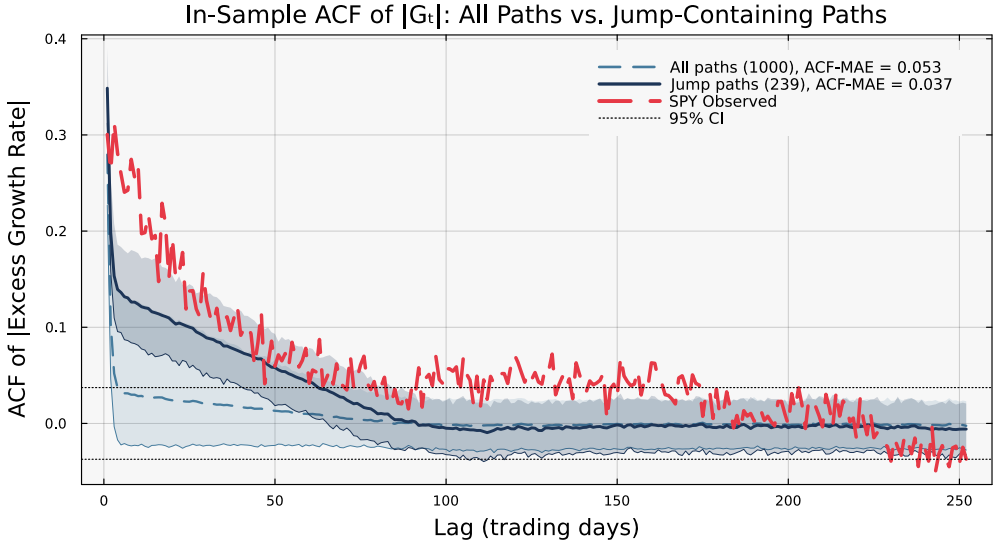


Fig. S3. In-sample ACF of $|G_t|$ for HMM-WJ: all paths vs. jump-containing paths. The mean ACF over all 1,000 paths (steel dashed) is diluted by the 76% of paths without jump events, producing an apparent failure to reproduce volatility clustering. Conditioning on the 239 paths that contain at least one Poisson jump (navy solid, with 10th–90th percentile band) recovers the persistent ACF decay observed in the SPY data (red dashed). ACF-MAE values are reported in the legend.

S7. Volatility clustering: all paths vs. jump-containing paths

The jump-duration mechanism produces volatility clustering only in the subset of simulated paths that contain at least one Poisson jump event. At the optimal $\epsilon^* = 10^{-4}$, approximately 24% of in-sample paths (239 out of 1,000) contained jumps; the remaining 76% followed pure Markov dynamics and exhibited no persistent ACF structure. When the ACF of $|G_t|$ was averaged over all 1,000 paths, the jump-path signal was diluted by the majority of non-jump paths, producing an ensemble-mean ACF that appeared to show little volatility clustering beyond lag 1. This dilution was an artifact of ensemble averaging, not a failure of the mechanism.

Figure S3 demonstrates this directly. The steel dashed curve shows the mean ACF averaged over all 1,000 paths: it drops to near zero by lag 5, which would suggest the model fails to reproduce volatility clustering. The navy solid curve shows the mean ACF averaged over only the 239 jump-containing paths: it tracks the observed SPY ACF (red dashed) closely across the full 252-lag horizon. The 10th–90th percentile band of the jump-path ensemble encompasses the observed ACF at most lags. This confirms that the jump mechanism does reproduce persistent volatility clustering within the paths it affects; the apparent gap in an all-paths average reflects the mixture composition of the ensemble rather than a structural limitation.

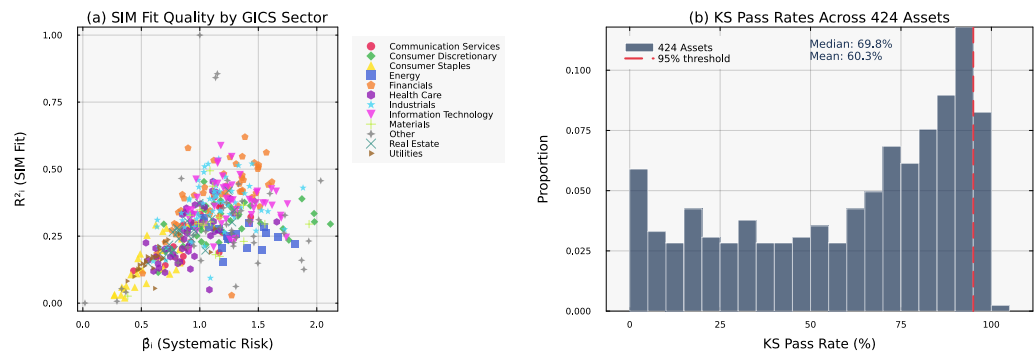


Fig. S4. Multi-asset extension via the Single-Index Model (SIM) across 424 S&P 500 constituents. Panel (a) shows the SIM regression fit quality (R^2 vs. $\hat{\beta}_i$) by GICS sector; assets with higher systematic risk exposure exhibit stronger factor model fit. Panel (b) shows the distribution of KS pass rates across the full asset universe (median 66.7%, mean 58.4%).

Table T3. Cross-sectional summary statistics for the Single-Index Model (SIM) extension across S&P 500 constituents. Individual asset paths are generated as $\hat{G}_{i,t} = \hat{\alpha}_i + \hat{\beta}_i G_{SPY,t} + \hat{\eta}_{i,t}$ where $G_{SPY,t}$ is drawn from HMM-WJ ($N = 100$) simulated paths and $\hat{\eta}_{i,t}$ is resampled from empirical residuals. KS pass rates computed over 1,000 paths at $\alpha = 0.05$.

Statistic	Mean	Median	5th pct	95th pct
<i>In-sample (424 assets, 2,766 trading days, 2014–2024)</i>				
KS pass rate (%)	58.4	66.7	2.5	96.0
SIM R^2	0.298	0.295	0.107	0.507
$\hat{\beta}$ (factor loading)	1.066	1.086	0.480	1.657
$\hat{\alpha}$ (intercept)	−0.0328	−0.0196	−0.2037	0.0964
<i>Out-of-sample (417 assets, 249 trading days, 2025)</i>				
KS pass rate (%)	82.1	91.8	23.8	99.0
SIM R^2	0.142	0.142	−0.169	0.479
$\hat{\beta}$ (factor loading)	1.064	1.084	0.482	1.660
$\hat{\alpha}$ (intercept)	−0.0320	−0.0186	−0.2035	0.0965

S8. Multi-asset SIM extension: detailed results

The Single-Index Model extension propagated HMM-WJ SPY paths to a 424-asset universe via the factor decomposition $\hat{G}_{i,t} = \hat{\alpha}_i + \hat{\beta}_i G_{SPY,t} + \hat{\eta}_{i,t}$, where idiosyncratic shocks were resampled from empirical residuals. Figure S4 shows the SIM fit quality by Global Industry Classification Standard (GICS) sector (panel a) and the distribution of KS pass rates across all 424 assets (panel b). Table T3 reports the cross-sectional summary statistics.

Table T4. Three-way comparison isolating the emission distribution and persistence mechanism for SPY ($N = 100$ for HMM-WJ, $K = 8$ for HSMM; 1,000 simulated paths, $\alpha = 0.05$). Standard errors in parentheses.

Metric	HSMM (Student- <i>t</i>)	HMM-WJ (Gaussian)	HMM-WJ (Student- <i>t</i>)
States	$K = 8$	$N = 100$	$N = 100$
Parameters	18	4	4
<i>In-sample (2,766 trading days, 2014–2024)</i>			
KS pass rate (%)	82.0 (1.2)	97.0 (0.5)	97.6 (0.5)
AD pass rate (%)	42.5 (1.6)	90.5 (0.9)	91.3 (0.9)
Excess kurtosis	4.8 (0.08)	5.5 (0.03)	7.6 (0.14)
ACF-MAE	0.059 (<0.001)	0.052 (<0.001)	0.052 (<0.001)
Wasserstein-1	0.176 (0.001)	0.101 (0.002)	0.101 (0.001)
Hellinger	0.113 (<0.001)	0.076 (<0.001)	0.075 (<0.001)
<i>Out-of-sample (249 trading days, 2025)</i>			
KS pass rate (%)	96.2 (0.6)	95.0 (0.7)	94.4 (0.7)
AD pass rate (%)	96.7 (0.6)	95.9 (0.6)	95.1 (0.7)
Excess kurtosis	4.1 (0.13)	4.9 (0.08)	6.1 (0.13)
ACF-MAE	0.042 (<0.001)	0.040 (<0.001)	0.039 (<0.001)
Wasserstein-1	0.287 (0.002)	0.275 (0.005)	0.282 (0.006)
Hellinger	0.239 (0.001)	0.212 (0.001)	0.210 (0.001)

S9. Emission distribution and semi-Markov comparison

Table T4 isolates the contributions of the emission distribution and the persistence mechanism by comparing three model variants on SPY: the HSMM baseline ($K = 8$ states, negative-binomial dwell times, Student-*t* emissions), HMM-WJ with the original Gaussian emissions ($N = 100$), and HMM-WJ with Student-*t* emissions ($N = 100$). The comparison confirms that HMM-WJ dominates the HSMM on distributional fidelity regardless of emission choice, and that the Student-*t* upgrade provides incremental improvements in tail reproduction (excess kurtosis closer to observed) and marginal fit (higher KS and AD pass rates).

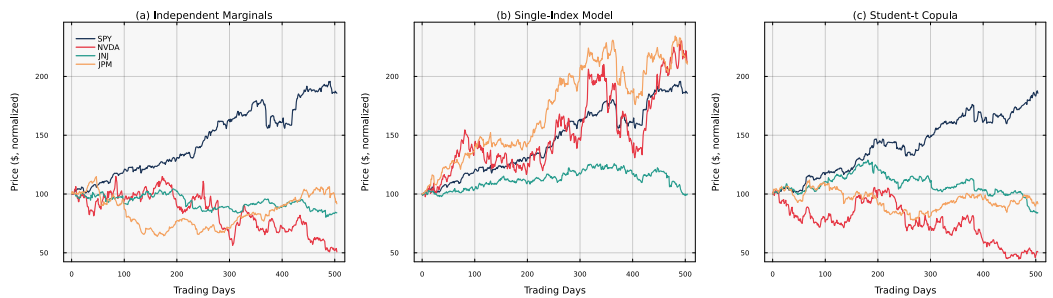


Fig. S5. Simulated price trajectories under three dependence specifications (all starting at \$100). (a) Independent HMM marginals: drawdowns and rallies occur at unrelated times. (b) Single-Index Model: partial co-movement via the market factor. (c) Student-t copula: crash episodes synchronize because the copula reorders each asset’s independently simulated values so that extreme days coincide. Panels (a) and (c) use the same growth rate values; only the temporal ordering differs.

S10. Price trajectory comparison

Figure S5 compares simulated price trajectories for the four-asset portfolio under three dependence specifications, demonstrating the visual effect of the copula reordering mechanism.

S11. Validation metric definitions

This section provides the formal definitions of the four validation metrics used in the main text to assess the distributional fidelity of synthetic paths.

Kolmogorov-Smirnov (KS) statistic. The two-sample KS statistic [37, 52] measures the maximum pointwise difference between two empirical CDFs:

$$D_{KS} = \sup_x |F_n(x) - F_m(x)| \quad (S12)$$

where F_n and F_m denote the empirical CDFs of the observed and simulated sequences of lengths n and m , respectively. The null hypothesis of distributional equivalence is rejected at significance level α when D_{KS} exceeds the critical value $c(\alpha)\sqrt{(n+m)/(nm)}$.

Anderson-Darling (AD) statistic. The AD statistic [2] places greater weight on the distributional tails than the KS test. The one-sample form measures deviation from a reference CDF F_0 :

$$A^2 = -n - \sum_{i=1}^n \frac{2i-1}{n} [\ln F_0(X_{(i)}) + \ln (1 - F_0(X_{(n+1-i)}))] \quad (S13)$$

where $X_{(1)} \leq \dots \leq X_{(n)}$ are the order statistics of the sample. In this study, we applied the two-sample variant, which replaces F_0 with the empirical CDF of the observed SPY series; critical values are obtained from the k -sample variant of [Anderson and Darling](#) as implemented in HypothesisTests.jl. The AD test is particularly suited for assessing tail-distributional fidelity, which is the primary concern for risk management applications.

Wasserstein-1 distance. The Wasserstein-1 distance measures the minimum work (distance $\times$ probability mass) required to transform one distribution into the other. For two equal-length empirical samples with order statistics $x_{(1)} \leq \dots \leq x_{(T)}$ and $y_{(1)} \leq \dots \leq y_{(T)}$:

$$W_1 = \frac{1}{T} \sum_{i=1}^T |x_{(i)} - y_{(i)}| \quad (S14)$$

W_1 is expressed in the same units as the data (daily excess growth rates) and equals zero only when the empirical distributions are identical. Because it is computed from sorted order statistics, the temporal ordering of observations is irrelevant.

Hellinger distance. The Hellinger distance measures distributional overlap on a bounded scale $[0, 1]$, with $H = 0$ indicating identical distributions and $H = 1$ indicating completely disjoint support. Given normalized histogram estimates $\{p_k\}_{k=1}^K$ and $\{q_k\}_{k=1}^K$ over a common K -bin grid:

$$H(P, Q) = \frac{1}{\sqrt{2}} \sqrt{\sum_{k=1}^K (\sqrt{p_k} - \sqrt{q_k})^2} \quad (S15)$$

Histograms were constructed on the common support of observed and simulated samples using $K = 50$ equal-width bins. The bounded scale makes H directly comparable across models and evaluation windows without rescaling.